Chaos, Solitons and Fractals 209 (2026) 118547

Contents lists available at ScienceDirect

Chaos, Solitons and Fractals

journal homepage: www.elsevier.com/locate/chaos

Homeostatic mechanisms in biological systems: Case studies in modeling copper regulation and self-immune recognition

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ARTICLE INFO

Dedicated to John Guckenheimer, Martin Golubitsky, Michael Field, Ian Stewart and Philip Holmes on the occasion of their 80th birthday.

Keywords:
Homeostasis
Input–output networks
Perfect adaptation

ABSTRACT

In this paper we investigate homeostatic mechanisms in two biologically motivated models: intracellular copper regulation and self immune recognition. The analysis is based on the notion of infinitesimal homeostasis for systems of ordinary differential equations and the formalism of input–output networks. We show that, unlike the examples of Reed et al. (2017), the obstruction of occurrence of infinitesimal homeostasis in both examples is due to the functional form of the differential equations being to far from the 'generic form'. More specifically, from a 'generic' or 'model-independent' point of view the models are expected to exhibit infinitesimal homeostasis, but due to functional form of the differential equations, forced by the modeling assumptions used to define the models, some of the 'expected generic behaviors' cannot occur. We first analyze both models from the 'model-independent' point of view and then show that in the specific models some 'expected generic behavior' cannot occur. We illustrate these phenomena with numerical simulations and give rigorous arguments in some cases.

1. Introduction

A system exhibits *homeostasis* if on change of an input variable $\mathcal{I}$ some observable $x_o(\mathcal{I})$ remains approximately constant. Many researchers have emphasized that homeostasis is an important phenomenon in biology. For example, the extensive work of Nijhout, Reed, Best and collaborators [1–5] consider biochemical networks associated with metabolic signaling pathways. Further examples include regulation of cell number and size [6], control of sleep [7], and expression level regulation in housekeeping genes [8].

Consider a dynamical system with input parameter $\mathcal{I}$ which varies over an open interval $]\mathcal{I}_1, \mathcal{I}_2[$. Suppose there is an output variable x_o such that for each $\mathcal{I} \in]\mathcal{I}_1, \mathcal{I}_2[$, the value $x_o(\mathcal{I})$ well-defined. In this situation, it is reasonable to say that the system would exhibit *homeostasis* if after changing the input variable $\mathcal{I}$, the value of the observable $x_o(\mathcal{I})$ remains approximately constant. There are two formulations often considered by researchers: (1) the strict condition of *perfect homeostasis*, where the observable $x_o(\mathcal{I})$ is required to be constant over a range of external stimuli; (2) the more general condition of *near-perfect homeostasis*, where the observable $x_o(\mathcal{I})$ is required to be within a narrow interval of values over a range of external stimuli.

Golubitsky and Stewart [9] proposed to employ methods from singularity theory to define the notion of *infinitesimal homeostasis*. According to this approach, a system exhibits *infinitesimal homeostasis* if $\frac{dx_o}{d\mathcal{I}}(\mathcal{I}_0) = 0$ for some input value $\mathcal{I}_0$, where x_o is the function that associates to each input parameter $\mathcal{I}_0$ a unique value of the observable x_o, called *input–output function*.

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https://doi.org/10.1016/j.chaos.2026.118547
Received 27 March 2026; Received in revised form 15 May 2026; Accepted 16 May 2026
Available online 23 May 2026

Reed et al. [10] considered four distinct small biochemical networks — also called *motif networks* — and determined which ones exhibit infinitesimal homeostasis: feedforward excitation, feedback product inhibition, the kinetic motif, and the parallel inhibition motif All of them occur in folate and methionine metabolism [1,4,4]. Interestingly, [10] showed that two of the motif networks exhibit infinitesimal homeostasis and that although the other two do not, they all exhibit near-perfect homeostasis (see also [11]).

Inspired by Reed et al. [10], Golubitsky and Wang [11] initiated a program to study homeostasis in the abstract setting of *input–output networks* (see [12], for the general theory). Here, the idea is to start with a directed graph with two distinguished nodes: the input node and the output node. Each node of the graph represents a (one-dimensional) state variable. Then one can attach to each such graph a set of admissible vector fields over the space of state variables, namely, the vector fields whose component functions respect the connections in the graph. Within this framework it is possible to formulate and prove results about the 'generic' occurrence of infinitesimal homeostasis in a given input–output network. This is a very common practice in the modern theory of nonlinear dynamical systems, namely, one seeks to characterize phenomena that (within the world of ODEs with a given network architecture) can occur typically, that is, for a non-empty open set of vector fields.

This perspective is called 'model-independent' approach (see Golubitsky and Stewart [13]). A model-independent property should normally remain valid if the model is slightly perturbed, within the relevant class. So model-independence implies some form of 'structural stability'. One of the main benefits of the model-independent approach is that when exact equations are not known in some area, it may still be possible to make 'model-independent predictions'. Another benefit is that the model-independent approach provides a catalog of the types of behavior that one should expect within a given class of models. If something different is observed, one can infer that something new is going on, ruling out entire classes of models, not just individual ones.

The model-independent approach seems to contrasts with the more common 'model-dependent' one, where specific model equations are solved, usually numerically. These two ways of thinking are not in competition: they complement each other. Specific models have the advantage to make connections with experimental tests. However, in most fields, like biology for instance, precise laws are often not known. Modelers therefore make assumptions about the functional form of the equations based on certain idealizations and approximations. More importantly, there is no assurance that the obtained specific model is 'generic'.

It is very common to find oneself in the following situation. The model-independent theory predicts that a given input–output network is expected to display infinitesimal homeostasis in several distinct ways. But numerical simulation of a specific model that is compatible with the network structure seems to suggest that there is no infinitesimal homeostasis at all, even under variation of other parameters.

This apparent discrepancy arises because, most of the time, the specific models are very far from being generic, even when considered as a parametrized family of models. The fact that the model-independent theory predicts certain behaviors to occur 'generically' does not mean that all admissible models should display that behavior. It means that, there are 'arbitrarily small perturbations' of any specific model that displays the expected behavior.

Let us consider an explicit example. *Feedback product inhibition* is probably one of the simplest and best known homeostatic mechanisms in biochemistry. In its simplest form, product inhibition means that the product of a biochemical chain inhibits one or more of the enzymes involved in its own synthesis. A realization of this mechanism is modeled by the following system of differential equations

$$\begin{aligned} \dot{x}_\iota &= \mathcal{I} - g_1(x_\iota) - f(x_\iota, x_o) \\ \dot{x}_\sigma &= f(x_\iota, x_o) - g_0(x_\sigma) - g_2(x_\sigma) \\ \dot{x}_o &= g_0(x_\sigma) - g_3(x_o) \end{aligned} \tag{1.1}$$

where f, g_i ($i = 0, 1, 2, 3$) are smooth functions. Although there are unspecified functions in the Eqs. (1.1), this system is not the most general admissible system in the sense of [11,12]. The model-independent analysis of product inhibition is carried out in [11, Sec. 2.3]

Typically, the functions g_i are defined on positive semi-axis, are linear and increasing. The function f is defined on the positive orthant and is positive. The actual kinetic formulas for the inhibitory function $f(x_\iota, x_o)$ have been extensively studied and depend on the details of the chemical binding of the substrate to one or more sites on the enzyme. One can impose inequality constraints on the function f in order to get similar behavior: $\frac{\partial f}{\partial x_\iota} > 0$ (more substrate, faster reaction) and $\frac{\partial f}{\partial x_o} < 0$ (higher substrate, more inhibition of the reaction).

Now, it can be shown (see [11]) that, for generic systems of the form (1.1) the input–output function x_o is well-defined for all $\mathcal{I} > 0$ and

$$\frac{dx_o}{d\mathcal{I}} = 0 \qquad \Longleftrightarrow \qquad \frac{\partial f}{\partial x_\iota} = 0$$

That is, the assumption that $\frac{\partial f}{\partial x_\iota} > 0$ precludes occurrence of infinitesimal homeostasis. Hence, any specific model such that the inhibitory function f respects the inequality constraints above will not exhibit infinitesimal homeostasis.

Nevertheless, it is shown in [10] that near-perfect homeostasis is possible in such systems if one chooses an f for which $\frac{\partial f}{\partial x_\iota} > 0$ is close to zero – such a choice is consistent with the biochemistry of feedback product inhibition. Furthermore, the condition that $\frac{\partial f}{\partial x_\iota} \approx 0$ implies that there are arbitrarily small perturbations of (1.1) that do exhibit infinitesimal homeostasis.

There is a second example in [10] exhibiting near-perfect homeostasis but not infinitesimal homeostasis, it is a network motif called *parallel inhibition* (a two-node network). Again, the conclusion that infinitesimal homeostasis cannot occur in this system, follows from an incompatibility of a biochemical condition, called *parallel inhibition hypotheses*, and the infinitesimal homeostasis

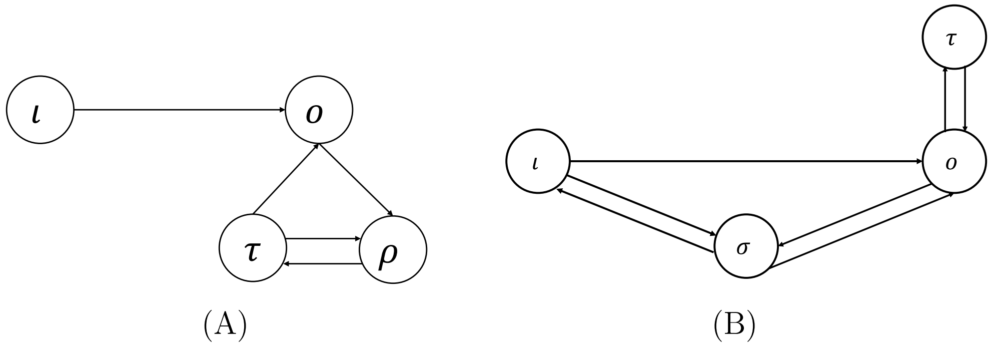

Fig. 1. Two abstract four-node input–output networks considered in this paper. (A) Intracellular copper regulation (core equivalent to network number 20 in [14]). (B) Self immune recognition (core equivalent to network number 18 in [14]).

condition that $\frac{dx_o}{d\mathcal{I}} = 0$. Therefore, in both examples the obstruction to the occurrence of infinitesimal homeostasis comes from additional modeling assumptions idealized from some characteristics of the phenomena being modeled.

In this paper we consider other mechanisms that may obstruct the occurrence of infinitesimal homeostasis in specific models, even though the existence of infinitesimal homeostasis is generically expected. We introduce and analyze two biologically motivated models: intracellular copper regulation and self immune recognition.

More specifically, we uncover two distinct phenomena here. In the first example (intracellular copper regulation) we show that an infinitesimal homeostasis point can disappear by variation of another parameter of the model. Moreover, the transition from occurrence to non-occurrence of an infinitesimal homeostasis point is connected with the existence of a limit point, that disappears together with the homeostasis point when this other parameter crosses a threshold value.

In the second example (self immune recognition) we show that the infinitesimal homeostasis point is forced to be at $+\infty$, that is the boundary of the domain of definition of the input–output-function. In this case, we have to extend the definition of infinitesimal homeostasis for this to make sense. We introduce the notion of 'asymptotic infinitesimal homeostasis' and give some criteria for its existence.

The two models that we explore in this paper can be represented by two four node networks shown in Fig. 1. Interestingly, the classification of four-node input–output networks (up to core equivalence, see Section 1.1 for the definitions of core equivalence) has been recently obtained by [14]. The examples we consider here correspond to the cases 20 and 18 of [14], respectively.

Structure of the paper. In the next two sections we analyze the two examples. For each example we give a biological motivation for the model and carry out in depth analyses of both models, first from the model-independent viewpoint and then by exploring the special properties of the specific model equations.

1.1. Dynamical formalism for homeostasis

Golubitsky and Stewart proposed a mathematical method for the study of homeostasis based on dynamical systems theory [9,15] (see the reviews [16,17]). In this framework, one consider a system of differential equations

$$\dot{X} = F(X, \mathcal{I}) \tag{1.2}$$

where $X = (x_1, \ldots, x_k) \in \mathbb{R}^k$ and parameter $\mathcal{I} \in \mathbb{R}$ represents the external input to the system.

Suppose that $(X^*, \mathcal{I}^*)$ is a linearly stable equilibrium of (1.2). By the implicit function theorem, there is a function $x(\mathcal{I})$ defined in a neighborhood of $\mathcal{I}^*$ such that $x(\mathcal{I}^*) = X^*$ and $F(x(\mathcal{I}), \mathcal{I}) \equiv 0$. The simplest case is when there is a variable, let us say x_k, whose output is of interest when $\mathcal{I}$ varies. Define the associated *input–output function* as $z(\mathcal{I}) = x_k(\mathcal{I})$. The input–output function allows one to formulate several definitions that capture the notion of homeostasis (see [9,15,18–20]).

Let $z(\mathcal{I})$ be the input–output function associated to a system of differential Eqs. (1.2) and the family of equilibria $x(\mathcal{I})$. We say that the corresponding system (1.2) exhibits

(a) *Perfect Homeostasis (Adaptation)* on the interval $(\mathcal{I}_1, \mathcal{I}_2)$ if

$$\frac{dz}{d\mathcal{I}}(\mathcal{I}) = 0 \qquad \text{for all } \mathcal{I} \in (\mathcal{I}_1, \mathcal{I}_2) \tag{1.3}$$

That is, z is constant on $(\mathcal{I}_1, \mathcal{I}_2)$.

(b) *Near-perfect Homeostasis (Adaptation)* relative to a *set point* $\mathcal{I}_{\rm sp}$ on the interval $(\mathcal{I}_1, \mathcal{I}_2)$ if, for a fixed δ,

$$|z(\mathcal{I}) - z(\mathcal{I}_{\rm sp})| \leqslant \delta \qquad \text{for all } \mathcal{I} \in (\mathcal{I}_1, \mathcal{I}_2) \tag{1.4}$$

That is, z stays within $z(\mathcal{I}_{\rm sp}) \pm \delta$ over $(\mathcal{I}_1, \mathcal{I}_2)$.

(c) *Infinitesimal Homeostasis* at the point $\mathcal{I}_c$ on the interval $(\mathcal{I}_1, \mathcal{I}_2)$ if

$$\frac{dz}{d\mathcal{I}}(\mathcal{I}_c) = 0 \tag{1.5}$$

That is, $\mathcal{I}_c$ is a *critical point* of z.

It is clear that perfect homeostasis implies near-perfect homeostasis, but the converse does not hold. Inspired by Reed et al. [2,21], Golubitsky and Stewart [9,15] introduced the notion of infinitesimal homeostasis that is intermediate between perfect and near-perfect homeostasis. It is obvious that perfect homeostasis implies infinitesimal homeostasis. On the other hand, it follows from Taylor's theorem that infinitesimal homeostasis implies near-perfect homeostasis in a neighborhood of $\mathcal{I}_0$. It is easy to see that the converse to both implications is not generally valid (see [10]). Moreover, the notion of infinitesimal homeostasis allows the tools from singularity theory to bear on the study of homeostasis.

When combined with coupled systems theory [22] the formalism of [9,15–17] becomes very effective in the analysis of model equations.

An *input–output network* is a network $\mathcal{G}$ with a distinguished *input node* ι, associated to the input parameter $\mathcal{I}$, one distinguished *output node* o, and N *regulatory nodes* $\rho = \{\rho_1, \dots, \rho_N\}$. The associated network systems of differential equations have the form

$$\begin{aligned} \dot{x}_\iota &= f_\iota(x_\iota, x_\rho, x_o, \mathcal{I}) \\ \dot{x}_\rho &= f_\rho(x_\iota, x_\rho, x_o) \\ \dot{x}_o &= f_o(x_\iota, x_\rho, x_o) \end{aligned} \tag{1.6}$$

where $\mathcal{I} \in \mathbb{R}$ is an *external input parameter* and $X = (x_\iota, x_\rho, x_o) \in \mathbb{R} \times \mathbb{R}^N \times \mathbb{R}$ is the vector of state variables associated to the network nodes. We write a vector field associated with the system (1.6) as

$$F(X, \mathcal{I}) = (f_\iota(X, \mathcal{I}), f_\rho(X), f_o(X))$$

and call it an *admissible vector filed* for the network $\mathcal{G}$. We say that $\mathcal{G}$ is the *influence network* of F. That is, each component f_j comprises the *influence* of the full state $X = (x_\iota, x_\rho, x_o)$ on the jth node state x_j (see [23, Sec. 2.2] for details).

Let f_{j,x_ℓ} denote the partial derivative of the jth node function f_j with respect to the ℓ^{th} node variable x_ℓ. We assume the following about the vector field F:

(a) The vector field F is smooth and has an asymptotically stable equilibrium at $(X^*, \mathcal{I}^*)$. Therefore, by the implicit function theorem, there is a function $x(\mathcal{I})$ defined in a neighborhood of $\mathcal{I}^*$ such that $x(\mathcal{I}^*) = X^*$ and $F(x(\mathcal{I}), \mathcal{I}) \equiv 0$.
(b) (Influence Network Condition) The partial derivative f_{j,x_ℓ} can be non-zero only if the network $\mathcal{G}$ has an arrow $\ell \to j$, otherwise $f_{j,x_\ell} \equiv 0$.
(c) (Genericity Condition) Only the input node coordinate function f_ι depends on the input parameter $\mathcal{I}$ and the partial derivative $f_{\iota,\mathcal{I}}$ generically satisfies

$$f_{\iota,\mathcal{I}} \neq 0. \tag{1.7}$$

The mapping $\mathcal{I} \mapsto x_o(\mathcal{I})$ is called the *input–output function* of the input–output network $\mathcal{G}$ (associated to the family of equilibria $x(\mathcal{I})$).

As noted previously [9,10,12,16], a straightforward application of Cramer's rule gives a simple formula for determining infinitesimal homeostasis points. Let J be the $(N+2) \times (N+2)$ Jacobian matrix of an admissible vector field $F = (f_\iota, f_\sigma, f_o)$, that is,

$$J = \begin{pmatrix} f_{\iota,x_\iota} & f_{\iota,x_\rho} & f_{\iota,x_o} \\ f_{\rho,x_\iota} & f_{\rho,x_\rho} & f_{\rho,x_o} \\ f_{o,x_\iota} & f_{o,x_\rho} & f_{o,x_o} \end{pmatrix} \tag{1.8}$$

The $(N+1) \times (N+1)$ matrix H obtained from J by dropping the last column and the first row is called *homeostasis matrix* of $\mathcal{G}$:

$$H = \begin{pmatrix} f_{\rho,x_\iota} & f_{\rho,x_\rho} \\ f_{o,x_\iota} & f_{o,x_\rho} \end{pmatrix} \tag{1.9}$$

In both Eqs. (1.8) and (1.9) partial derivatives f_{ℓ,x_j} are evaluated at $\big(x(\mathcal{I}), \mathcal{I}\big)$.

Lemma 1.1. *The input–output function $x_o(\mathcal{I})$ of an input–output network $\mathcal{G}$ satisfies*

$$x_o'(\mathcal{I}) = \pm f_{\iota,\mathcal{I}} \frac{\det(H)}{\det(J)} \tag{1.10}$$

Here, x_o' is the derivative of x_o with respect to $\mathcal{I}$ and $\det(J)$, $\det(H)$ are evaluated at $\big(x(\mathcal{I}), \mathcal{I}\big)$. Hence, $\mathcal{I}_0$ is a point of infinitesimal homeostasis if and only if

$$\det(H) = 0 \tag{1.11}$$

at the equilibrium $\big(x(\mathcal{I}_0), \mathcal{I}_0\big)$.

Proof. See [12,17]. □

The determinant $\det(H)$ is called *homeostasis determinant*. Two networks input–output $\mathcal{G}_1$ and $\mathcal{G}_2$ are called *core equivalent* if they have the same homeostasis determinant (up to re-labeling the regulatory nodes).

Moreover, one can uniquely decompose $\det(H)$, seen as a multivariate polynomial in the partial derivatives f_{j,x_ℓ}, into irreducible factors

$$\det(H) = \det(B_1) \dots \det(B_k)$$

where each B_η is a diagonal block of the upper triangular form of H, obtained by permuting rows and columns [12]. Let us denote the irreducible factors of $\det(H)$ as

$$h_\eta(\mathcal{I}) \equiv \det B_\eta(X(\mathcal{I}), \mathcal{I})$$

We say that B_η (or h_η) generically induces homeostasis at $\mathcal{I}_0$ if $h_\eta(\mathcal{I}_0) = 0$ and $h_\xi(\mathcal{I}_0) \neq 0$ for all $\xi = \eta$.

2. Intracellular copper regulation

2.1. Brief review of copper regulation

Copper is an inorganic element essential to many physiological process, including neurotransmission, gastrointestinal uptake, lactation, transport to the developing brain and growth. However, its concentration must be tightly regulated, as intracellular copper excess is associated to cellular damage and protein folding disorders [24,25].

In addition to cytosolic copper concentration, copper in intra-mitochondrial space must be also strictly regulated, as it is paramount for the function of copper dependent enzymes, but it may cause oxidative stress in excessive levels [26].

Copper in the external medium enters the cell by a copper transport protein CTR1, that transports copper across the plasma membrane with high affinity and specificity. In the cytosol, copper is rapidly incorporated to glutatione, from where it is ligated to metallochaperones, as ATOX1, CCS and COX17. The enzyme ATOX1 is associated to the copper 'secretory pathway', while CCS and COX17 are enrolled in incorporating copper in the mitochondrial enzymes SOD1 and COX [24,25].

The enzyme ATOX1 takes the cytosolic copper to the Cu-ATPases ATP7A and ATP7B, which use ATP to pump copper ions to vesicles of the trans-Golgi network, where copper will be incorporated in Cu-dependent enzymes and secreted. This is called the 'secretory pathway' and it is responsible for decreasing the cytosolic copper concentration. However, when cytosolic copper levels are too low, ATP7A and ATP7B take copper from the trans-Golgi network and give it to ATOX1, leading to an increase in cytosolic copper concentration [27].

The functions governed by copper homeostasis are primarily executed by the copper-transporting ATPases known as ATP7A and ATP7B. The enzyme ATP7A is a transmembrane protein located throughout the body, except for the liver, with two essential roles in copper homeostasis: (i) transporting copper across cell membranes in both directions (regulating absorption of copper only in the small intestines, and excreting excessive intracellular copper, in all tissues), therefore, aiming at the maintenance of intracellular copper concentrations (both cytosolic and mitochondrial); and (ii) participating as a cofactor in the activating mechanisms of copper-dependent enzymes, critical for the structure and function of bone, skin, hair, blood vessels, and the nervous system [28,29]. On the other hand, the ATP7B transmembrane protein is located primarily in liver cells, but also in the brain, and carry out similar functions as ATP7A: regulating intracellular copper concentrations by releasing copper into bile and plasma, and co-activating copper-dependent enzymes in the Golgi apparatus [24,29].

Let us briefly expand on the physiological implications of defective copper regulation. Starting with anomalies in the ATP7B gene. They can generate a sole disorder known as Wilson disease (WD), in which dysfunctional ATP7B proteins implicate WD carriers to accumulate abnormal levels of copper in the liver and in the brain. As a result, clinical features comprise neurological, hepatic, psychiatric and skeletal abnormalities, as well as renal tubular dysfunction and hemolytic anemia. The prognosis in WD is generally favorable given that current therapeutic approaches prevent or attenuate most of the symptoms. Its chronic nature, however, implies that treatment interruption results in potentially fatal liver damage [30].

Differently, variations in the ATP7A gene result in dysfunctional ATP7A proteins that cause three separate illnesses: Menkes disease, a severe early-onset neurodegenerative condition in which carriers usually die by 3 years of age [31]; occipital horn syndrome, a connective disorder with typical skeleton deformations which is also clinically resembling to Menkes disease, while less aggressive in its neurological manifestation [32]; and a recently found distal motor neuropathy, marked by frequent onset at adulthood and with no apparent signs of copper metabolic abnormalities, although still poorly studied [33,34].

2.2. Mathematical model

A simple model for the intracellular copper regulation represented by Fig. 2 can be obtained as follows. Consider the concentration of copper in four compartments: extracellular copper $[\mathrm{Cu_{ext}}]$, cytosolic copper $[\mathrm{Cu_{cyt}}]$, trans-Golgi stored copper $[\mathrm{Cu_{TG}}]$ and mitochondrial copper $[\mathrm{Cu_{mit}}]$; and their regulation by three metallochaperones: ATOX1, CCS and COX17. The relations among these components is represented by the input–output network shown in Fig. 3(A). Recently, a more detailed model for intracellular copper regulation has been proposed in [35]. See also [17] for an updated version of this model.

In this network, the extracellular copper concentration $[\mathrm{Cu_{ext}}]$ is the input node and mitochondrial copper concentration $[\mathrm{Cu_{mit}}]$ is the output node. In addition, we consider the concentration of copper stored in the vesicles of the trans-Golgi network $[\mathrm{Cu_{TG}}]$

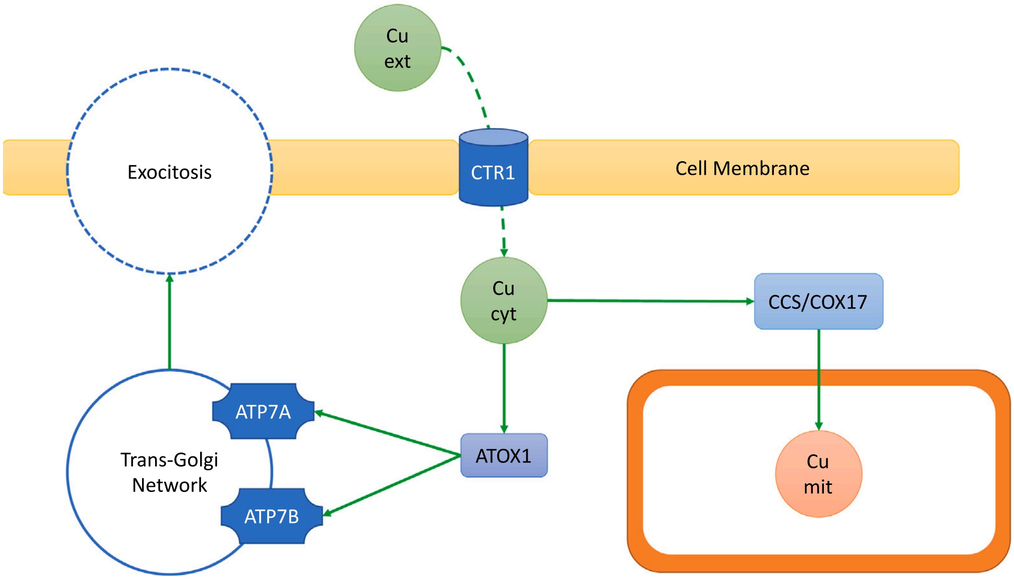

Fig. 2. A model for intracellular copper regulation. Here, Cu_ext: extracellular copper; Cu_cyt: cytosolic copper; Cu_mit: mitochondrial copper.

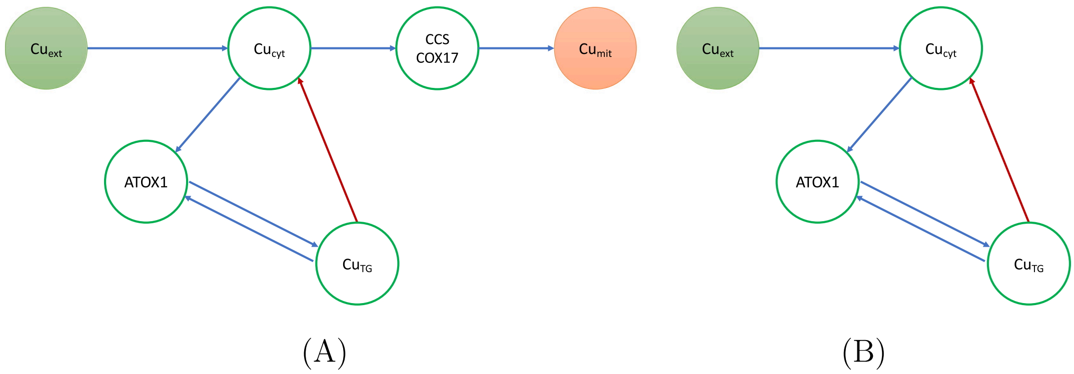

Fig. 3. Input–output network for the intracellular copper regulation model. Blue arrows indicate positive stimulus (activation) and red arrows indicate negative stimulus (inhibition). (A) Full network. (B) Reduced network. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

and the corresponding enzyme [ATOX1] that pumps the stored copper back into the cytosol. The input parameter $\mathcal{I}$ represents the abundance of extracellular copper. Observe that, in order for [Cu_{mit}] to be homeostatic, it is enough for [Cu_{cyt}] to be homeostatic. Hence, we can further simplify the input–output network of Fig. 3(A) to the *reduced* input–output network shown in Fig. 3(B). Note that, as an abstract input–output network, the network shown in Fig. 3(B) is exactly the network shown in Fig. 1(A).

A system of model equations for the dynamics of intracellular copper represented by the reduced input–output network, shown in Fig. 3(B), is given below. To facilitate notation, we denote the concentrations of Cu_{ext}, Cu_{cyt}, Cu_{TG} and ATOX1, respectively, by x_ι, x_o, x_ρ and x_τ.

The model equations associated with the network in Fig. 3(B) is given by

$$\begin{aligned}
\dot{x}_\iota &= \mathcal{I} - k_0 x_\iota \\
\dot{x}_\tau &= f k_1 x_o - k_3 x_\tau - w_2(x_\rho - x_\tau)H(x_\tau) \\
\dot{x}_\rho &= g k_3 x_\tau - k_4 x_\rho + w_2(x_\rho - x_\tau)H(x_\tau) \\
\dot{x}_o &= \frac{k_0}{N} x_\iota - k_1 x_o(1 + w_1 x_o) - k_2 G(x_\rho)
\end{aligned} \tag{2.12}$$

Here, $\mathcal{I} \geqslant 0$, the coefficients N, w_1, w_2, k_0, k_1, k_2, k_3, k_4 are positive parameters, $f, g \in (0, 1]$ and G and H are Hill/Michaelis–Menten Functions (for $x \geq 0$):

$$G(x) = \frac{x^2}{1 + x^2} \qquad \text{and} \qquad H(x) = \frac{x}{1 + x} \tag{2.13}$$

Since the quantities x_ι, x_τ, x_ρ, x_o are concentrations we only consider solutions in the positive closed orthant $\{(x_\iota, x_\tau, x_\rho, x_o) \in \mathbb{R}^4 : x_j \geqslant 0\}$.

2.3. Model-independent analysis

The general form of the ODEs associated to the abstract input–output network from Fig. 1(A) is

$$\begin{aligned} \dot{x}_\iota &= f_\iota(x_\iota, \mathcal{I}) \\ \dot{x}_\tau &= f_\tau(x_\tau, x_\rho, x_o) \\ \dot{x}_\rho &= f_\rho(x_\tau, x_\rho) \\ \dot{x}_o &= f_o(x_\iota, x_\rho, x_o) \end{aligned} \tag{2.14}$$

The jacobian matrix J of (2.12) at a generic equilibrium point is

$$J = \begin{pmatrix} f_{\iota,x_\iota} & 0 & 0 & 0 \\ 0 & f_{\tau,x_\tau} & f_{\tau,x_\rho} & f_{\tau,x_o} \\ 0 & f_{\rho,x_\tau} & f_{\rho,x_\rho} & 0 \\ f_{o,x_\iota} & 0 & f_{o,x_\rho} & f_{o,x_o} \end{pmatrix} \tag{2.15}$$

So, the homeostasis determinant is given by

$$\det(H) = f_{o,x_\iota}\,(f_{\tau,x_\tau} f_{\rho,x_\rho} - f_{\tau,x_\rho} f_{\rho,x_\tau}) \tag{2.16}$$

Hence, the abstract network supports *Haldane homeostasis*, that is, $f_{o,x_\iota}(X(\mathcal{I}_0), \mathcal{I}_0) = 0$ for some $\mathcal{I}_0$, and *appendage homeostasis*, that is, $(f_{\tau,x_\tau} f_{\rho,x_\rho} - f_{\tau,x_\rho} f_{\rho,x_\tau})(X(\mathcal{I}_0), \mathcal{I}_0) = 0$ for some $\mathcal{I}_0$.

2.3.1. Appendage homeostasis and negative feedback loops

In control theory, there is the notion of 'feedback loop', which basically has two types: 'positive' and 'negative'. The terms positive and negative refer to how the dynamical system respond to disturbances. The term 'negative feedback' is usually employed when the dynamical system reacts to disturbances in a way that decreases the effect of those disturbances. Whilst, 'positive feedback' usually applies when the increase in some signal leads to a situation in which a certain quantity is further increased through its dynamics [36].

In the context of input–output networks we can formulate these concepts using the notion of cycles (closed paths) in directed graphs.

Definition 2.1. Let $\mathcal{G}$ e an input–output network.

(a) A *cycle* C in a $\mathcal{G}$ is a *closed path*, that is. $C = \{\sigma_1 \to \sigma_2 \to \cdots \sigma_n \to \sigma_1\}$ of $\mathcal{G}$ such that $\sigma_i \neq \sigma_j$ if $i \neq j$.
(b) A cycle C is called a*positive or excitatory cycle* if the product of the linearized coupling strengths along C is positive. Similarly, C is called a *negative or inhibitory cycle* if the product of the linearized coupling strengths along C is negative.

We say that $\mathcal{G}$ contains a *feedback loop at the output node* if $\mathcal{G}$ contains a cycle starting and ending at node o.

Let us consider the general admissible system (2.14). The abstract network displays a cycle $C = \{o \to \rho \to \tau \to o\}$ that is a feedback loop at the out put node according to the definition 2.1. Moreover, the subnetwork induced by the regulatory nodes ρ and τ forms what is called an *appendage subnetwork*. That is, it is the subnetwork associated with the irreducible factor $\det(H)$ that induces appendage homeostasis (see [12]), namely,

$$h_2 = f_{\tau,x_\tau} f_{\rho,x_\rho} - f_{\tau,x_\rho} f_{\rho,x_\tau}$$

Now, from Eq. (2.15) it is clear that the eigenvalues of J are f_{ι,x_ι} and the eigenvalues of the matrix

$$J_{(1,1)} = \begin{pmatrix} f_{\tau,x_\tau} & f_{\tau,x_\rho} & f_{\tau,x_o} \\ f_{\rho,x_\tau} & f_{\rho,x_\rho} & 0 \\ 0 & f_{o,x_\rho} & f_{o,x_o} \end{pmatrix} \tag{2.17}$$

The characteristic polynomial of $J_{(1,1)}$ is

$$p_{J_{(1,1)}}(\lambda) = \begin{vmatrix} f_{\tau,x_\tau} - \lambda & f_{\tau,x_\rho} & f_{\tau,x_o} \\ f_{\rho,x_\tau} & f_{\rho,x_\rho} - \lambda & 0 \\ 0 & f_{o,x_\rho} & f_{o,x_o} - \lambda \end{vmatrix}$$

which means that

$$\begin{aligned} p_{J_{(1,1)}}(\lambda) = &-\lambda^3 + \lambda^2(f_{\tau,x_\tau} + f_{\rho,x_\rho} + f_{o,x_o}) \\ &-\lambda(f_{\tau,x_\tau} f_{\rho,x_\rho} - f_{\tau,x_\rho} f_{\rho,x_\tau} + f_{\tau,x_\tau} f_{o,x_o} + f_{\rho,x_\rho} f_{o,x_o}) \\ &+ (f_{\tau,x_\tau} f_{\rho,x_\rho} - f_{\tau,x_\rho} f_{\rho,x_\tau}) f_{o,x_o} + f_{\tau,x_o} f_{\rho,x_\tau} f_{o,x_\rho} \end{aligned}$$

Assuming that the system is expected to exhibit appendage homeostasis, that is, $f_{\tau,x_\tau} f_{\rho,x_\rho} - f_{\tau,x_\rho} f_{\rho,x_\tau} = 0$, we can simplify the characteristic polynomial to

$$p_{J_{(1,1)}}(\lambda) = -\lambda^3 + \lambda^2(f_{\tau,x_\tau} + f_{\rho,x_\rho} + f_{o,x_o}) - \lambda(f_{\tau,x_\tau} f_{o,x_o} + f_{\rho,x_\rho} f_{o,x_o}) + f_{\rho,x_\tau} f_{\tau,x_o} f_{o,x_\rho}$$

That is, the product of the three eigenvalues of $J_{(1,1)}$ is $f_{\rho,x_\tau} f_{\tau,x_o} f_{o,x_\rho}$, which is the product of the linearized couplings along the feedback loop C.

By the assumption that the system exhibits appendage homeostasis, we are also assuming that the equilibrium is linearly stable, that is, all the eigenvalues of $J_{(1,1)}$ have negative real part. Thus, if all roots of the characteristic polynomial are real, than all of them should be negative, which implies that $f_{\rho,x_\tau} f_{\tau,x_o} f_{o,x_\rho} < 0$. On the other hand, if one of the eigenvalues is not real, its complex conjugate is also an eigenvalue, as all the coefficients of $p_{J_{(1,1)}}$ are real. The product of a non-null complex number by its conjugate is positive, and therefore the product of the three eigenvalues in that case must also be negative and so, $f_{\rho,x_\tau} f_{\tau,x_o} f_{o,x_\rho} < 0$, as well. Hence, we have shown the following.

Proposition 2.1. *Suppose that the general admissible system (2.14) exhibits appendage homeostasis for some point $\mathcal{I}_0$. If the corresponding equilibrium $X(\mathcal{I}_0)$ is linearly stable then the feedback loop $C = \{o \to \rho \to \tau \to o\}$ is negative.*

That is, stability together with appendage homeostasis forces negative feedback!

2.3.2. Higher order derivatives of the input–output function

One of the main advantages of the notion of infinitesimal homeostasis is that one can bring ideas from singularity theory to bear on the analysis of input–output functions. For instance, the first higher order derivative of the input–output function that does not vanish at a critical point 'determines' the local structure in a neighborhood of the critical point, see [9].

The simplest case is called *simple homeostasis* and occurs when

$$x_o'(\mathcal{I}_0) = 0 \qquad \text{and} \qquad x_o''(\mathcal{I}_0) \neq 0$$

The next case is called *chair homeostasis* [21] and occurs when

$$x_o'(\mathcal{I}_0) = 0 \qquad \text{and} \qquad x_o''(\mathcal{I}_0) = 0 \qquad \text{and} \qquad x_o'''(\mathcal{I}_0) \neq 0$$

These condition are called *defining conditions* of the singularity type. In the first case, the input–output function is topologically equivalent to a quadratic function near $\mathcal{I}_0$, namely

$$x_o(\mathcal{I}) = \pm(\mathcal{I} - \mathcal{I}_0)^2$$

This type of singularity is called *non-degenerate* or *Morse* [37]. In the second case, the input–output function is topologically equivalent to a one parameter family of cubic functions near $\mathcal{I}_0$, namely

$$x_o(\mathcal{I}, \alpha) = \pm(\mathcal{I} - \mathcal{I}_0)^3 + \alpha(\mathcal{I} - \mathcal{I}_0)$$

where the 'chair' critical point occurs when $\alpha = 0$ [37]. The main difference between these two situations is that the first is structurally stable, while the second is not when $\alpha = 0$ [38,39]. Nevertheless, the one parameter family of functions in the second case, is such that, when $\alpha \neq 0$, it gives all small perturbations (up to changes of coordinates) of the case $\alpha = 0$ [37]. Hence, knowing the higher order derivatives of x_o at an infinitesimal point $\mathcal{I}_0$ can help one to fully characterize to local topological structure of the input–output function near $\mathcal{I}_0$.

Using the factorization of $\det(H)$ into irreducible factors it is possible to write a formula for the second derivative of the input–output function, at a point of infinitesimal homeostasis $\mathcal{I}_0$,

$$x_o''(\mathcal{I}) = \pm \frac{f_{\iota,\mathcal{I}}}{\det(J)} \frac{d}{d\mathcal{I}} \det(H)$$

where

$$\det(H) = f_{o,x_\iota} (f_{\tau,x_\tau} f_{\rho,x_\rho} - f_{\tau,x_\rho} f_{\rho,x_\tau})$$

Let us write the homeostasis determinant as $\det(H) = h_1\, h_2$ with

$$h_1 = f_{o,x_\iota} \qquad \text{and} \qquad h_2 = f_{\tau,x_\tau} f_{\rho,x_\rho} - f_{\tau,x_\rho} f_{\rho,x_\tau}$$

then

$$\left.\frac{d}{d\mathcal{I}} \det(H)\right|_{\mathcal{I}_0} = h_1'(\mathcal{I}_0)\, h_2(\mathcal{I}_0) + h_1(\mathcal{I}_0)\, h_2'(\mathcal{I}_0)$$

Hence, if $h_1(\mathcal{I}_0) \neq 0$ and $h_2(\mathcal{I}_0) = 0$ we have that the defining conditions of the singularity type is completely determined by the higher derivatives of h_2. In this regard, the following proposition is useful.

Proposition 2.2. *Let H be the homeostasis matrix of the general system (2.14). Then, at a point of infinitesimal homeostasis $\mathcal{I}_0$, we have that*

$$h_1' = \frac{-f_{\iota,\mathcal{I}}}{\det(J)}\left(f_{o,x_o}h_2 + \det(J_{(1,1)})\right)\frac{\partial h_1}{\partial x_\iota} \tag{2.18}$$

where $J_{(1,1)}$ is defined in (2.17) and

$$h_2' = \frac{-f_{\iota,\mathcal{I}}}{\det(J)} f_{o,x_\iota} f_{\tau,x_o}\left(f_{\rho,x_\rho}\frac{\partial h_2}{\partial x_\tau} - f_{\rho,x_\tau}\frac{\partial h_2}{\partial x_\rho}\right) \tag{2.19}$$

Proof. Since $h_j(\mathcal{I}) \equiv h_j(X(\mathcal{I}),\mathcal{I})$, we have that

$$\frac{d}{d\mathcal{I}}h_j = \frac{\partial h_j}{\partial x_\iota}\frac{dx_\iota}{d\mathcal{I}} + \frac{\partial h_j}{\partial x_\tau}\frac{dx_\tau}{d\mathcal{I}} + \frac{\partial h_j}{\partial x_\rho}\frac{dx_\rho}{d\mathcal{I}} + \frac{\partial h_j}{\partial x_o}\frac{dx_o}{d\mathcal{I}} \tag{2.20}$$

As we are evaluating (2.20) at an infinitesimal homeostasis point, $\frac{d}{d\mathcal{I}}x_o = 0$. Furthermore, in the case of h_1 the expression does not explicitly depends on x_τ and x_ρ, and in the case of h_2 the expression does not explicitly depends on x_ι.

Therefore, we may simplify (2.20), obtaining

$$\frac{d}{d\mathcal{I}}h_1 = \frac{\partial h_1}{\partial x_\iota}\frac{dx_\iota}{d\mathcal{I}} \tag{2.21}$$

and

$$\frac{d}{d\mathcal{I}}h_2 = \frac{\partial h_2}{\partial x_\tau}\frac{dx_\tau}{d\mathcal{I}} + \frac{\partial h_2}{\partial x_\rho}\frac{dx_\rho}{d\mathcal{I}} \tag{2.22}$$

Now we need to compute $\frac{dx_\iota}{d\mathcal{I}}$, $\frac{dx_\tau}{d\mathcal{I}}$ and $\frac{dx_\rho}{d\mathcal{I}}$. In order to perform this, we shall use a strategy analogous to the one used to obtain the homeostasis matrix. That is, since J is the Jacobian at the homeostasis point and $f_{\iota,\mathcal{I}} = 1$, then the following linear system is satisfied

$$J\begin{pmatrix} x_\iota' \\ x_\tau' \\ x_\rho' \\ x_o' \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \\ 0 \\ 0 \end{pmatrix} \tag{2.23}$$

where $'$ indicates derivative with respect to $\mathcal{I}$. As the equilibrium is linearly stable, $\det(J) \neq 0$, and therefore we may apply Cramer's rule to compute $\frac{dx_\iota}{d\mathcal{I}}$, $\frac{dx_\tau}{d\mathcal{I}}$ and $\frac{dx_\rho}{d\mathcal{I}}$. Therefore, we can write

$$\frac{dx_\iota}{d\mathcal{I}} = \frac{\det(H_\iota)}{\det(J)} \quad \text{and} \quad \frac{dx_\tau}{d\mathcal{I}} = \frac{\det(H_\tau)}{\det(J)} \quad \text{and} \quad \frac{dx_\rho}{d\mathcal{I}} = \frac{\det(H_\rho)}{\det(J)} \tag{2.24}$$

where

$$\det H_\iota = \begin{vmatrix} -f_{\iota,\mathcal{I}} & 0 & 0 & 0 \\ 0 & f_{\tau,x_\tau} & f_{\tau,x_\rho} & f_{\tau,x_o} \\ 0 & f_{\rho,x_\tau} & f_{\rho,x_\rho} & 0 \\ 0 & 0 & f_{o,x_\rho} & f_{o,x_o} \end{vmatrix} \tag{2.25}$$

and

$$\det H_\tau = \begin{vmatrix} f_{\iota,x_\iota} & -f_{\iota,\mathcal{I}} & 0 & 0 \\ 0 & 0 & f_{\tau,x_\rho} & f_{\tau,x_o} \\ 0 & 0 & f_{\rho,x_\rho} & 0 \\ f_{o,x_\iota} & 0 & f_{o,x_\rho} & f_{o,x_o} \end{vmatrix} \tag{2.26}$$

and

$$\det H_\rho = \begin{vmatrix} f_{\iota,x_\iota} & 0 & -f_{\iota,\mathcal{I}} & 0 \\ 0 & f_{\tau,x_\tau} & 0 & f_{\tau,x_o} \\ 0 & f_{\rho,x_\tau} & 0 & 0 \\ f_{o,x_\iota} & 0 & 0 & f_{o,x_o} \end{vmatrix} \tag{2.27}$$

Hence, using that $\det(J_{(1,1)}) = f_{\rho,x_\tau} f_{\tau,x_o} f_{o,x_\rho}$, we obtain

$$\det(H_\iota) = -f_{\iota,\mathcal{I}}\left(f_{o,x_o}h_2 + \det(J_{(1,1)})\right) \tag{2.28}$$

and

$$\det(H_\tau) = -f_{\iota,\mathcal{I}} f_{o,x_\iota} f_{\tau,x_o} f_{\rho,x_\rho} \quad \text{and} \quad \det(H_\rho) = f_{\iota,\mathcal{I}} f_{o,x_\iota} f_{\tau,x_o} f_{\rho,x_\tau} \tag{2.29}$$

By combining (2.24) and (2.28) to (2.22), we obtain (2.18) and by combining (2.24) and (2.29) to (2.22), we obtain (2.19). □

We end this section by noting that the same method used in the proof of (2.19) can be employed to compute the second order derivative, or higher order derivatives, of the input–output function of any input–output network.

2.4. Analysis of the specific model

Now let us turn to the analysis of the actual model Eqs. (2.12). Consider the jacobian matrix (2.15) of the general admissible system (2.14). Substituting the model Eqs. (2.12) into (2.15) gives

$$J = \begin{pmatrix} -k_0 & 0 & 0 & 0 \\ 0 & -k_3 + w_2\frac{x_\tau^2+2x_\tau-x_\rho}{(1+x_\tau)^2} & -w_2 H(x_\tau) & f k_1 \\ 0 & g k_3 - w_2\frac{x_\tau^2+2x_\tau-x_\rho}{(1+x_\tau)^2} & -k_4 + w_2 H(x_\tau) & 0 \\ \frac{k_0}{N} & 0 & -k_2 G_{x_\rho}(x_\rho) & -k_1(1+2w_1 x_o) \end{pmatrix}$$

Note that, for $\mathcal{I}=0$, the point $X(0)=(0,0,0,0)$ is a solution and the jacobian at $X(0)=(0,0,0,0)$ is (recall that $G_{x_\rho}(0)=0$)

$$J\Big|_{\mathcal{I}=0} = \begin{pmatrix} -k_0 & 0 & 0 & 0 \\ 0 & -k_3 & 0 & f k_1 \\ 0 & g k_3 & -k_4 & 0 \\ \frac{k_0}{N} & 0 & 0 & -k_1 \end{pmatrix}$$

and so $X(0)=(0,0,0,0)$ is always a stable equilibrium. Hence, the input–output function is well-defined near $\mathcal{I}=0$. Although $\mathcal{I}$ only makes biological sense when it is non-negative, the input–output function is defined for $\mathcal{I}$ negative, as well.

The homeostasis determinant for the model equations is given by

$$\det(H) = \frac{k_0}{N}\left(k_4 \left(k_3 - w_2 \frac{x_\tau^2 + 2x_\tau - x_\rho}{(1+x_\tau)^2} \right) + k_3 w_2 (g-1) H(x_\tau) \right) \tag{2.30}$$

and for $\mathcal{I}=0$ we have that

$$\det(H)\Big|_{\mathcal{I}=0} = \frac{k_0 k_3 k_4}{N} \neq 0$$

That, is $\mathcal{I}=0$ is not a point of infinitesimal homeostasis. Additionally, we have that, for all $\mathcal{I}$,

$$f_{o,x_\iota} = \frac{k_0}{N} \neq 0$$

Therefore, if the system exhibits homeostasis, it exhibits only *appendage homeostasis*.

In general, the input–output function depends on all the other parameters of the model equations, that is, x_o, is a function of $\mathcal{I}$ and the coefficients N, w_1, w_2, k_0, k_1, k_2, k_3, k_4, f, g. Thus, it is very difficult to find an explicit expression of x_o, for arbitrary parameter values.

Usually, the best one can do is to fix the values of the all the parameters and numerically compute the input–output function, as a function of $\mathcal{I}$. Indeed, by this method we can show that the model equations exhibits infinitesimal homeostasis for certain parameter values.

In Fig. 4 we show a plot of the input–output of the model Eqs. (2.12), obtained by numerical continuation of the equilibrium $X(0)=(0,0,0,0)$, as a function of $\mathcal{I}$. Here, the input–output function is given by the stable branch (red curve) starting at $(x_o,\mathcal{I})=(0,0)$ and ending at the limit point that occurs when $\mathcal{I}_{\text{sn}} \approx 5.2$. Moreover, one can see that there is a critical point at $\mathcal{I}_0 \approx 4.7$. This an infinitesimal homeostasis point given by the occurrence of *appendage homeostasis*. The choice of the parameter values is just for the sake of illustration, we do not know if these values are physiologically realistic

The graph shown in Figs. 4(A) and (B) suggests that, for an open set of parameter values, the system exhibits *simple homeostasis*, that is, the second derivative of input–output function at $\mathcal{I}_0$ is non-zero.

If this turns out to be true then we expect that this configuration is persistent under small variation of the parameter values. This follows from that fact that a simple homeostasis point is a Morse singularity, which is structurally stable [38,39]. This can be easily verified numerically.

Nevertheless, this can breakdown under large variation of parameters. Indeed, as shown in Figs. 5(A) and (B), variation of the parameter w_2 form $w_2 = 0.5$ to $w_2 = 0.340$ and $w_2 = 0.339$, respectively, does not change the topological type of the singularity. The effect of decreasing w_2 from 0.5 towards 0.339 is that the critical point $\mathcal{I}_0$ and the limit point $\mathcal{I}_{\text{sn}}$ get closer to each other. Upon crossing the threshold value $w_2^* \approx 0.339$ there is a transition in the topology of the input–output function. The critical point $\mathcal{I}_0$ and the limit point $\mathcal{I}_{\text{sn}}$ disappear (apparently the 'collide' and 'annihilate' each other) and the function becomes monotonically increasing. Furthermore, it seems that the input–output function becomes defined on the whole positive semi-axis, $D=[0,+\infty)$.

The analysis of the model equations for the copper intra-cellular regulation raises several interesting questions. The most basic one, motivated by Fig. 4 is the following. Given that the model Eqs. (2.12) is only capable of exhibiting appendage homeostasis, if it exhibits infinitesimal homeostasis at all, is it the case that the singularity type is that of simple homeostasis?

We can use Proposition 2.2 to determine the singularity type of the appendage homeostasis whenever it occurs in the model Eqs. (2.12).

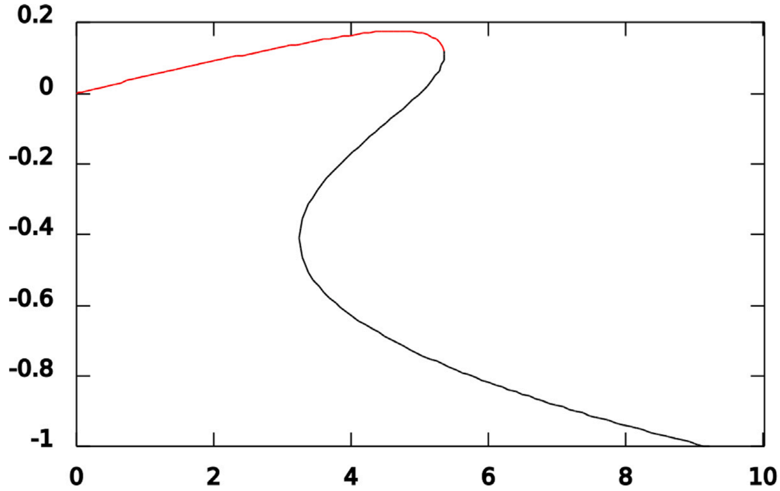

Fig. 4. Projection onto the $(x_o, \mathcal{I})$-plane of the numerical continuation of the equilibrium $X(\mathcal{I})$ generated by Xpp-Auto [40]. The y-axis is x_o and the x-axis is $\mathcal{I}$. The input–output function is given by the stable branch (red curve) starting at $\mathcal{I} = 0$ and ending at a limit point $\mathcal{I}_{sn} \approx 5.2$. The black curve is the unstable branch, which seems to be unbounded from below. Infinitesimal homeostasis occurs at the critical point $\mathcal{I}_0 \approx 4.7$. Parameter values are: $N = 10$, $f = 0.5$, $g = 0.05$, $w_1 = 1$, $w_2 = 0.5$, $k_0 = 10$, $k_1 = 2$, $k_2 = 1$, $k_3 = 0.5$, $k_4 = 1$. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

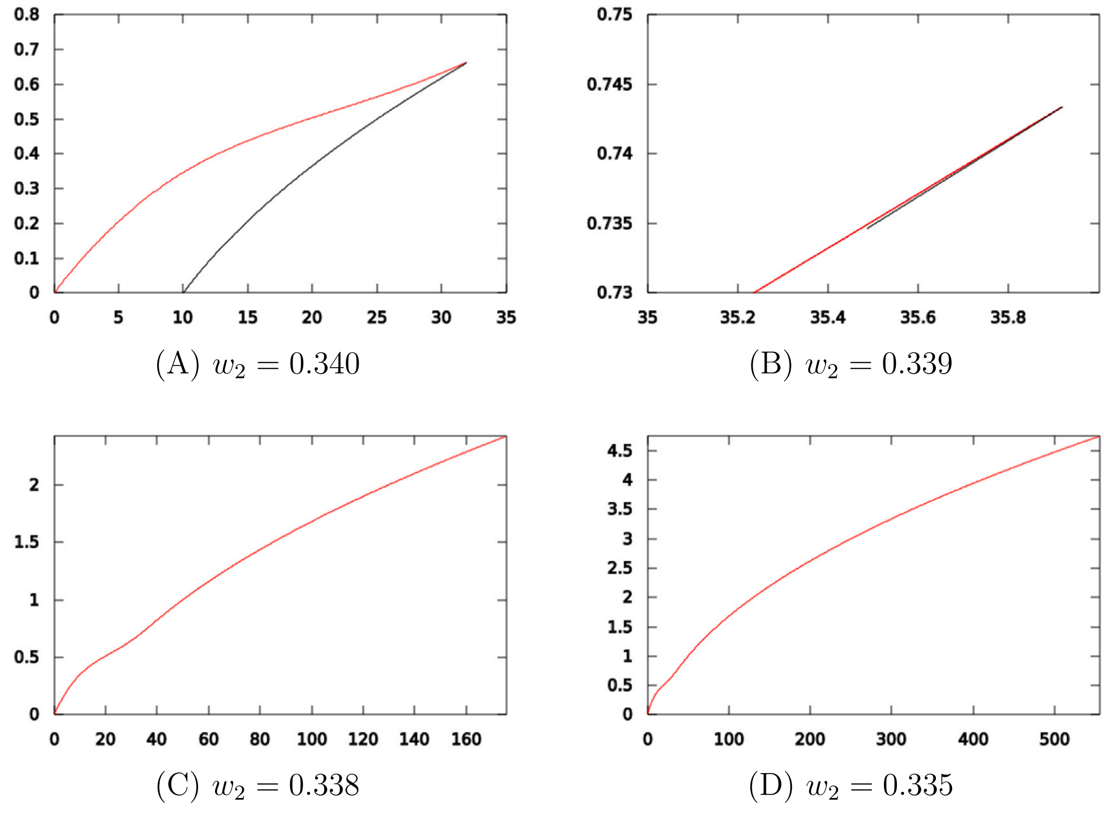

Fig. 5. Projection onto the $(x_o, \mathcal{I})$-plane of the numerical continuation of the equilibrium $X(\mathcal{I})$ generated by Xpp-Auto [40]. The y-axis is x_o and the x-axis is $\mathcal{I}$. The input–output function is given by the stable branch (red curve). Each panel correspond to different values of the parameter w_2, the other parameter values are fixed at the same values as in Fig. 4. In panels (A) and (B) w_2 decreases from 0.340 towards 0.339. In these two cases the input–output function still exhibits infinitesimal homeostasis at $\mathcal{I}_0$ and a limit point $\mathcal{I}_{sn}$. In panels (C) and (D) w_2 decreases from 0.338 to 0.335. In these two cases the input–output function becomes monotonically increasing and does not exhibits infinitesimal homeostasis. Note, that the scale of the x-axis is much larger in (C) and (D). The threshold value where the transition occurs is $w_2^* \approx 0.339$. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Proposition 2.3. *Consider the model Eqs. (2.12) associated with the input–output network shown in Fig. 3(B), with fixed but arbitrary parameter values. Suppose that the corresponding input–output function $x_o(\mathcal{I})$ exhibits infinitesimal homeostasis at some $\mathcal{I}_0 > 0$. Then the infinitesimal homeostasis is of appendage type, that is, it is induced by the factor $h_2(\mathcal{I}_0) = 0$ and it is a simple homeostasis, that is, $h_2'(\mathcal{I}_0) \neq 0$*

Proof. The first assertion, that $h_2(\mathcal{I}_0) = 0$, follows from that fact that $h_1 \neq 0$ for all $\mathcal{I}$, hence if $x_o'(\mathcal{I}_0) = 0$ it must be because $h_2(\mathcal{I}_0) = 0$. To prove the second assertion we use formula (2.19) from Proposition 2.2 with the explicit functions of the model Eqs. (2.12). The function h_2 is given by

$$h_2 = k_4\left(k_3 - w_2\frac{x_\tau^2 + 2x_\tau - x_\rho}{(1+x_\tau)^2}\right) + k_3 w_2 (g-1)\left(\frac{x_\tau}{1+x_\tau}\right)$$

Hence, its partial derivatives are

$$\begin{aligned}\frac{\partial h_2}{\partial x_\tau} &= \frac{(g-1)k_3 w_2}{(1+x_\tau)^2} - \frac{2k_4 w_2 (x_\rho + 1)}{(1+x_\tau)^3} \\ \frac{\partial h_2}{\partial x_\rho} &= \frac{k_4 w_2}{(1+x_\tau)^2}\end{aligned} \tag{2.31}$$

We also need the following partial derivatives

$$f_{\rho,x_\tau} = g k_3 - w_2\frac{x_\tau^2 + 2x_\tau - x_\rho}{(1+x_\tau)^2} \text{ and } f_{\rho,x_\rho} = -k_4 + w_2\frac{x_\tau}{1+x_\tau} \tag{2.32}$$

By assumption, we have that $\det J \neq 0$ and $f_{o,x_\iota} \neq 0$ at $\mathcal{I}_0$. Moreover, by Proposition 2.1, the feedback loop $o \to \rho \to \tau \to o$ is a negative feedback loop, which means that $f_{\tau,x_o} \neq 0$. Therefore, $h_2'(\mathcal{I}_0) = 0$ if and only if

$$f_{\rho,x_\tau}\frac{\partial h_2}{\partial x_\rho} - f_{\rho,x_\rho}\frac{\partial h_2}{\partial x_\tau} = 0 \tag{2.33}$$

at $\mathcal{I}_0$. Substituting (2.31) and (2.32) into (2.33), we obtain

$$\begin{aligned}&\left[g k_3 - w_2\frac{x_\tau^2 + 2x_\tau - x_\rho}{(1+x_\tau)^2}\right]\cdot\frac{k_4 w_2}{(1+x_\tau)^2} \\ &\quad + \left(k_4 - w_2\frac{x_\tau}{1+x_\tau}\right)\cdot\left[\frac{(g-1)k_3 w_2}{(1+x_\tau)^2} - \frac{2k_4 w_2(x_\rho+1)}{(1+x_\tau)^3}\right] = 0\end{aligned} \tag{2.34}$$

Rearranging the terms gives

$$\begin{aligned}&\frac{(2g-1)k_3 k_4 w_2}{(1+x_\tau)^2} - \frac{w_2}{(1+x_\tau)^2}\left[\frac{(g-1)k_3 w_2 x_\tau}{(1+x_\tau)} - \frac{k_4 w_2 (x_\tau^2 + 2x_\tau - x_\rho)}{(1+x_\tau)^2}\right] \\ &\quad - \frac{2k_4 w_2^2 (x_\tau^2 + 2x_\tau - x_\rho)}{(1+x_\tau)^4} - \frac{2k_4^2 w_2 (x_\rho + 1)}{(1+x_\tau)^3} + \frac{2k_4 w_2^2 (x_\rho+1)x_\tau}{(1+x_\tau)^4} = 0\end{aligned} \tag{2.35}$$

Since the system exhibits appendage homeostasis, from (2.30), we obtain

$$h_2 = 0 \quad \Leftrightarrow \quad \frac{(g-1)k_3 w_2 x_\tau}{(1+x_\tau)} - \frac{k_4 w_2 (x_\tau^2 + 2x_\tau - x_\rho)}{(1+x_\tau)^2} = -k_3 k_4 \tag{2.36}$$

Form (2.36) and (2.34), we obtain

$$\frac{2g k_3 k_4 w_2}{(1+x_\tau)^2} - \frac{2k_4 w_2^2 (x_\tau^2 + 2x_\tau - x_\rho)}{(1+x_\tau)^4} - \frac{2k_4^2 w_2 (x_\rho+1)}{(1+x_\tau)^3} + \frac{2k_4 w_2^2 (x_\rho+1)x_\tau}{(1+x_\tau)^4} = 0 \tag{2.37}$$

and so

$$\begin{aligned}&\frac{2k_4 w_2}{(1+x_\tau)^2}\left[g k_3 - \frac{w_2(x_\tau^2 + 2x_\tau - x_\rho)}{(1+x_\tau)^2}\right] \\ &\quad + \frac{2k_4 w_2 (1+x_\rho)}{(1+x_\tau)^3}\left[-k_4 + \frac{w_2 x_\tau}{(1+x_\tau)}\right] = 0\end{aligned} \tag{2.38}$$

Now substituting (2.32) into (2.38) gives

$$\begin{aligned}&\frac{2k_4 w_2}{(1+x_\tau)^2} f_{\rho,x_\tau} + \frac{2k_4 w_2 (1+x_\rho)}{(1+x_\tau)^3} f_{\rho,x_\rho} = \\ &\frac{2k_4 w_2}{(1+x_\tau)^3}\left[(1+x_\tau) f_{\rho,x_\tau} + (1+x_\rho) f_{\rho,x_\rho}\right] = 0\end{aligned} \tag{2.39}$$

From Eq. (2.39), it follows that $\frac{2k_4 w_2}{(1+x_\tau)^3} \neq 0$, and therefore in $h_2'(\mathcal{I}_0) = 0$ is equivalent to

$$(1+x_\tau) f_{\rho,x_\tau} + (1+x_\rho) f_{\rho,x_\rho} = 0 \tag{2.40}$$

Thus, we have shown that the simultaneous validity of $h_2(\mathcal{I}_0) = 0$ and $h_2'(\mathcal{I}_0) = 0$ is equivalent to the following equations being simultaneously satisfied

$$\begin{aligned}(1+x_\tau)f_{\rho,x_\tau} + (1+x_\rho)f_{\rho,x_\rho} &= 0\\ -f_{\tau,x_\rho}f_{\rho,x_\tau} + f_{\tau,x_\tau}f_{\rho,x_\rho} &= 0\end{aligned} \tag{2.41}$$

By treating these equations as an homogeneous linear system in variables f_{ρ,x_τ} and f_{ρ,x_ρ} and using that f_{ρ,x_τ}, we obtain

$$\begin{vmatrix}(1+x_\tau) & (1+x_\rho)\\ -f_{\tau,x_\rho} & f_{\tau,x_\tau}\end{vmatrix} = 0 \Rightarrow (1+x_\tau)f_{\tau,x_\tau} + (1+x_\rho)f_{\tau,x_\rho} = 0 \tag{2.42}$$

Now, recall the following partial derivatives

$$f_{\tau,x_\tau} = -k_3 + w_2\frac{x_\tau^2 + 2x_\tau - x_\rho}{(1+x_\tau)^2} \quad \text{and} \quad f_{\tau,x_\rho} = -w_2\frac{x_\tau}{(1+x_\tau)} \tag{2.43}$$

Substituting (2.43) into (2.42) gives

$$\begin{aligned}(1+x_\tau)f_{\tau,x_\tau} + (1+x_\rho)f_{\tau,x_\rho} &=\\ -k_3(1+x_\tau) + w_2\frac{x_\tau^2 + 2x_\tau - x_\rho}{(1+x_\tau)} - w_2\frac{x_\tau(1+x_\rho)}{(1+x_\tau)} &= 0\end{aligned} \tag{2.44}$$

That is

$$w_2\frac{x_\tau^2 + 2x_\tau - x_\rho}{(1+x_\tau)} - w_2\frac{x_\tau(1+x_\rho)}{(1+x_\tau)} = k_3(1+x_\tau) \tag{2.45}$$

From (2.32) and (2.40) we get

$$\begin{aligned}(1+x_\tau)f_{\rho,x_\tau} + (1+x_\rho)f_{\rho,x_\rho} &=\\ gk_3(1+x_\tau) - \left[w_2\frac{x_\tau^2 + 2x_\tau - x_\rho}{(1+x_\tau)} - w_2\frac{x_\tau(1+x_\rho)}{(1+x_\tau)}\right] - k_4(1+x_\rho) &= 0\end{aligned} \tag{2.46}$$

Finally, combining (2.45) and (2.46) gives

$$\begin{aligned}gk_3(1+x_\tau) - k_3(1+x_\tau) - k_4(1+x_\rho) &=\\ (g-1)k_3(1+x_\tau) - k_4(1+x_\rho) &= 0\end{aligned} \tag{2.47}$$

Since we assume that the equilibrium $X(\mathcal{I})$ is positive for all $\mathcal{I}$, it follows that $1 + x_\rho > 0$ and $1 + x_\tau > 0$. Moreover, as $0 < g \leq 1$, the expression on the left handed side of (2.47) is strictly negative, which is a contradiction. Therefore, we should have $h_2'(\mathcal{I}_0) \neq 0$. □

Combining Proposition 2.3 with the classification of structurally stable singularities of scalar-valued functions [38,39] gives the following.

Corollary 2.4. *Consider the input–output function x_o of the model Eqs. (2.12), as depending on all parameters of the model. Then there is an open set of parameters such that x_o exhibits infinitesimal homeostasis of appendage type and another open set of parameters such that x_o is monotonic over its domain.*

It is an interesting question to determine which of the two parameter regimes given by Corollary 2.4, if any, approximate what happen in real cells. We can, however, compare with other models to see if there is some convergence of predictions. In fact, [35, Fig. 6] shows a plot of the input–output function obtained from their model (they use a different terminology, calling it *homeostatic effectiveness*). They show two curves: red for the 'wild type' and green for the 'mutant type'; with the green curve being much flatter than the red curve. Moreover, they report a slope of just 0.70 μM (micromolar) of cystolic copper x_ι per μM change in the external copper $\mathcal{I}$, over the interval $[10, 50]$.

Both curves in [35, Fig. 6] are very similar to our curves from Fig. 5, panels (C) and (D), that is, the regime where x_o is monotonic. In fact, the curve in Fig. 5(D) is very flat, with an average slope of 0.009 over the interval $[0, 500]$. We can use the formula for the derivative of the input–output function $x_o(\mathcal{I})$ from Lemma 1.1 to make Xpp-Auto numerically compute $x_o'(\mathcal{I})$, see Fig. 6. From the plot of x_o' it looks like $x_o' \to 0$, very slowly, as $\mathcal{I} \to +\infty$ and, nevertheless, from Fig. 5(D), it seems that x_o is unbounded on $D = [0, +\infty)$.

3. Self immune recognition

3.1. Brief review of immune recognition

The immune system has a paramount role in mammalian physiology: it must combat any strange body and infection, and, at the same time, it must discriminate between which elements belong to the organism and which not in order to avoid autoimmunity, something know in the literature as self and non-self recognition. Although specificity of receptors expressed by immune cells is a major mechanism that explains the capacity of discrimination between self and non-self components, conventional T lymphocytes in tissues may still be erroneously activated leading to autoimmunity and cell injury [41].

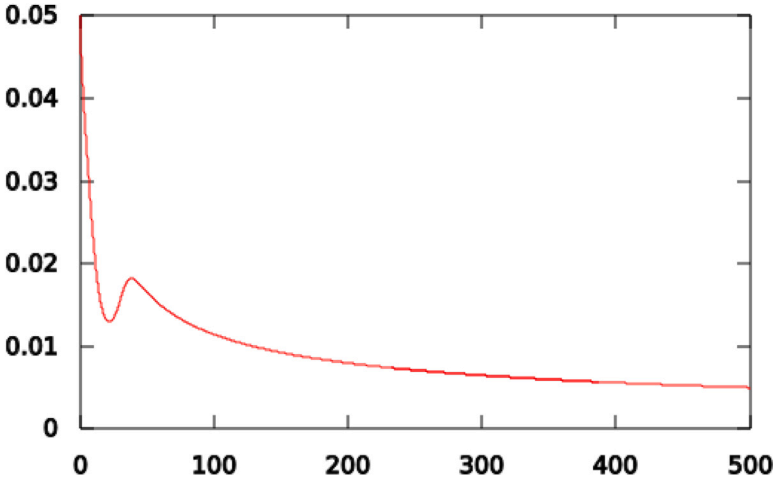

Fig. 6. Plot of the derivative $x_o'(\mathcal{I})$, corresponding to $x_o(\mathcal{I})$ in Fig. 5(D), obtained by numerical continuation of the equilibrium $X(\mathcal{I})$ generated by Xpp-Auto [40]. Here, $x_o'(0) = 0.05$ and $x_o'(500) \approx 0.005$. Parameter values are: $N = 10$, $f = 0.5$, $g = 0.05$, $w_1 = 1$, $w_2 = 0.335$, $k_0 = 10$, $k_1 = 2$, $k_2 = 1$, $k_3 = 0.5$, $k_4 = 1$.

Consider the three main immune cells that are present in tissues: antigen-presenting cells (APCs), responsible for initiating the immune response, conventional T lymphocytes (Tconv), which are the main cells responsible for a specific response against non-self pathogens, and regulatory T lymphocytes (Treg), which suppress Tconv activity [41–43]. Usually, the immune response starts when APCs take digested antigens and couple them to MHC molecules expressed in APCs surface [43]. This enables the recognition of the antigen by Tconv. When activated, Tconv cells synthesize interleukin-2 (IL-2), which stimulates both Tconv and Treg cells. On the other hand, Treg interacts to Tconv, particularly with autoreactive Tconv, supressing their activity [41,43]. The importance of Treg cells may be exemplified by the fact that patients with pathogenic variants in the gene *FOXP3*, leading to Treg cells dysfunction develop an autoimmune syndrome called IPEX (Immune dysregulation, polyendocrinopathy, enteropathy, X-linked syndrome) [44].

3.2. Mathematical model

We introduce a simple model, inspired by Khailaie et al. [43], that considers the situation where the only existing antigen is '*self*'.

The model is given by the interplay between four compartments: APCs (x_τ), Tconv (x_o), IL-2 (x_σ) and Treg (x_ι) and the input parameter $\mathcal{I}$ represents antigen stimulation of the concentration of Treg. The model equations are

$$\begin{aligned}
\dot{x}_\iota &= a x_\iota x_\sigma - b x_\iota + \mathcal{I} \\
\dot{x}_\sigma &= c x_o - d x_\sigma (x_o + x_\iota) - e x_\sigma \\
\dot{x}_\tau &= -b x_\tau + f \frac{x_\tau}{x_\tau + g} + h x_\tau x_o + j \\
\dot{x}_o &= a x_\sigma x_o - b x_o - l x_\iota x_o + h x_\tau x_o + j
\end{aligned} \tag{3.48}$$

Here, $\mathcal{I} \geqslant 0$, and the coefficients $a, b, c, d, e, f, g, h, j, l$ are positive parameters. Since the quantities x_ι, x_σ, x_τ, x_o are concentrations, we only consider solutions in the positive closed orthant $\{(x_\iota, x_\sigma, x_\tau, x_o) \in \mathbb{R}^4 : x_j \geqslant 0\}$.

The equations for $\dot{x}_\iota$ and $\dot{x}_\sigma$ in (3.48) are the same as the equations for $\dot{R}$ and $\dot{I}$ in [43, Eq. (13)]. The equation for $\dot{x}_o$ is obtained from the equation for $\dot{T}$ in [43, Eq. (13)] by substituting the steady state value of Treg activation in the equations and omitting the quadratic term (cT^2). The equation for $\dot{x}_\tau$ is obtained from the equation for $\dot{N}$ in [43, Eq. (10)] by substituting the steady state value of Treg activation in the equation and adding the APC self-activation given by a Michaelis–Menten mechamism.

Associating the labels ι, σ, τ and o to the nodes of an input–output network one can check that the model Eqs. (3.48) are admissible (as an influence network, see Section 1.1) for the input–output network shown in Fig. 1(B).

Observe that in model Eqs. (3.48) the parameter j (the basal level of APCs and Tconv) could be considered as a second 'input parameter'. Indeed, it is possible to extend the theory of [12] to input–output networks with multiple inputs [23] and apply to the model Eqs. (3.48). However, to keep the analysis as simple as possible, i.e., in the context of single-input single-output networks, will consider $j > 0$ as a fixed parameter and only treat $\mathcal{I}$ as an external disturbance. Although, in the formal analysis of the model below, we will consider the extreme case where $j = 0$ and compare with the case $j > 0$.

3.3. Model-independent analysis

The general form of the ODEs associated to the abstract input–output network from Fig. 1(B) is

$$\begin{aligned}\dot{x}_\iota &= f_\iota(x_\iota, x_\sigma, \mathcal{I})\\ \dot{x}_\sigma &= f_\sigma(x_\iota, x_\sigma, x_o)\\ \dot{x}_\tau &= f_\tau(x_\tau, x_o)\\ \dot{x}_o &= f_o(x_\iota, x_\sigma, x_\tau, x_o)\end{aligned} \tag{3.49}$$

The jacobian matrix J of (2.12) at a generic equilibrium point is

$$J = \begin{pmatrix} f_{\iota,x_\iota} & f_{\iota,x_\sigma} & 0 & 0\\ f_{\sigma,x_\iota} & f_{\sigma,x_\sigma} & 0 & f_{\sigma,x_o}\\ 0 & 0 & f_{\tau,x_\tau} & f_{\tau,x_o}\\ f_{o,x_\iota} & f_{o,x_\sigma} & f_{o,x_\tau} & f_{o,x_o}\end{pmatrix} \tag{3.50}$$

So, the homeostasis determinant is given by

$$\det(H) = f_{\tau,x_\tau}\,(f_{o,x_\sigma} f_{\sigma,x_\iota} - f_{o,x_\iota} f_{\sigma,x_\sigma}) \tag{3.51}$$

Hence, the abstract network supports *null-degradation homeostasis*, that is, $f_{\tau,x_\tau}(X(\mathcal{I}_0), \mathcal{I}_0) = 0$ for some $\mathcal{I}_0$, and *structural homeostasis*, that is, $(f_{o,x_\sigma} f_{\sigma,x_\iota} - f_{o,x_\iota} f_{\sigma,x_\sigma})(X(\mathcal{I}_0), \mathcal{I}_0) = 0$ for some $\mathcal{I}_0$.

Note that, unlike the matrix (2.15), which has only one non-zero term (f_{τ,x_o}) on the first row/last column, besides f_{ι,x_ι} and f_{o,x_o}, matrix (3.50) have three: f_{ι,x_σ}, f_{σ,x_o} and f_{τ,x_o}. These terms correspond to the arrows $\sigma \to \iota$, $o \to \sigma$ and $o \to \tau$, respectively. Moreover, the only common term, f_{τ,x_o}, is the one related to the feedback loop and the equilibrium's stability in both cases (see [17, Ex. 2.10]).

3.4. Analysis of the specific model

Now let us turn to the analysis of the actual model Eqs. (3.48). Substituting the vector field components of model Eqs. (3.48) into (3.50) gives

$$J = \begin{pmatrix} ax_\sigma - b & ax_\iota & 0 & 0\\ -dx_\sigma & -d(x_o + x_\iota) - e & 0 & c - dx_\sigma\\ 0 & 0 & -b + \frac{fg}{(x_\tau+g)^2} + hx_o & hx_\tau\\ -lx_o & ax_o & hx_o & ax_\sigma - b - lx_\iota + hx_\tau\end{pmatrix}$$

Let us first assume that $j = 0$ (for the purpose of theoretical analysis), then for $\mathcal{I} = 0$, the point $X(0) = (0, 0, 0, 0)$ is a solution and the jacobian at $X(0) = (0, 0, 0, 0)$ is

$$J\Big|_{\mathcal{I}=0} = \begin{pmatrix} -b & 0 & 0 & 0\\ 0 & -d-e & 0 & c\\ 0 & 0 & -b+f/g & 0\\ 0 & 0 & 0 & -b\end{pmatrix}$$

Thus, $X(0) = (0, 0, 0, 0)$ is a stable equilibrium if and only if $b > f/g$. Therefore, under this condition, the input–output function is well-defined near $\mathcal{I} = 0$. Although $\mathcal{I}$ only makes biological sense when it is non-negative, the input–output function is defined for $\mathcal{I}$ negative, as well.

The homeostasis determinant for the model equations is given by

$$\det(H) = -x_o\,(adx_\sigma + ldx_\iota + ldx_o + de)\left(-b + \frac{fg}{(x_\tau + g)^2} + hx_o\right) \tag{3.52}$$

and for $j = 0$ and $\mathcal{I} = 0$ we have that $f_{\tau,x_\tau} = -b + f/g$ and

$$\det(H)\Big|_{\mathcal{I}=0} = 0$$

That, is $\mathcal{I} = 0$ is a point of infinitesimal homeostasis, that is caused by the vanishing of *structural homeostasis* factor. Even more is true, since from the equation of $\dot{x}_o$ in (3.48), assuming $j = 0$ and $\mathcal{I} > 0$, we have that

$$0 = x_o\,(ax_\sigma - b - lx_\iota - hx_\tau) \tag{3.53}$$

That is, $x_o \equiv 0$ and so the system exhibits *perfect homeostasis* (see Fig. 7(A)).

When $j > 0$ is small, linear stability of the equilibrium $X(\mathcal{I}, j)$ at $(\mathcal{I}, j) = (0, 0)$ implies linear stability for small $(\mathcal{I}, j)$ near $(0, 0)$.

However, when $j > 0$ the homeostasis determinant is no longer zero at $\mathcal{I} = 0$. Furthermore, numerical simulation suggests that, when $j > 0$, $\det(H) \neq 0$ for all $\mathcal{I} > 0$ and $x_o \to 0$ when $\mathcal{I} \to +\infty$ (see Fig. 7(B)).

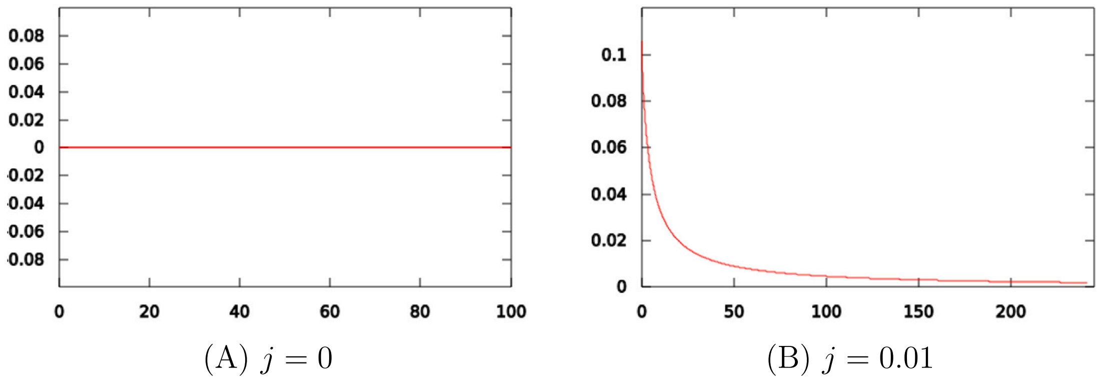

Fig. 7. Projection onto the $(x_o, \mathcal{I})$-plane of the numerical continuation of the equilibrium $X(\mathcal{I})$ generated by Xpp-Auto [40]. The y-axis is x_o and the x-axis is $\mathcal{I}$. Panel (A) shows the input–output function x_o when $j = 0$, which is explicitly given by $x_o \equiv 0$. Panel (B) shows the input–output function when $j = 0.01$. Here, x_o does not exhibit infinitesimal homeostasis at any point. The remaining parameters are: $a = 0.5$, $b = 1.0$, $c = 0.5$, $d = 1.0$, $e = 0.5$, $f = 0.05$, $g = 0.3$, $h = 0.1$, $l = 0.2$.

Let us show that, indeed, $\det H \neq 0$, when $j > 0$, for all $\mathcal{I} > 0$. First, note that $x_o \neq 0$ at equilibrium (this follows from the equation for $\dot{x}_o$ in (3.48)). The condition for occurrence of structural homeostasis is that the following expression vanishes

$$f_{o,x_\sigma} f_{\sigma,x_\iota} - f_{o,x_\iota} f_{\sigma,x_\sigma} = -x_o[adx_\sigma + ldx_\iota + ldx_o + de] \tag{3.54}$$

Since the equilibrium point $(x_\iota, x_\sigma, x_\tau, x_o)$ is assumed to be in the positive orthant, we conclude that the system does exhibits *structural homeostasis*. Finally, we need to show that system does not exhibits *null-degradation homeostasis*, as well. Suppose that $f_{\tau,x_\tau} = 0$ at equilibrium, then

$$x_o = \frac{b}{h} - \frac{fg}{h(x_\tau + g)^2} \tag{3.55}$$

Form (3.55) and the equation for $\dot{x}_\tau$ we get

$$fx_\tau^2 + j(x_\tau + g)^2 = 0 \tag{3.56}$$

As the last equality cannot hold, the system does not exhibit *null degradation homeostasis*. Therefore, the system does not exhibit infinitesimal homeostasis.

3.4.1. Infinitesimal homeostasis at infinity

As suggested by numerical simulation, Fig. 7(B), when $j > 0$, the input–output function seems to defined on the whole positive semi-axis $D = (0, +\infty)$ and does not exhibit infinitesimal homeostasis. Moreover, when $\mathcal{I} \to +\infty$ it suggests that $x_o \to 0$ and $x_o' \to 0$, as well. This last condition is called *asymptotic infinitesimal homeostasis* (see Definition A.1 in Appendix A).

Proposition 3.1. *Let x_o be the input–output function of associated to the model Eq. (3.48), with $j \neq 0$ and $\mathcal{I} \geqslant 0$. Suppose that x_o is defined over the positive semi-axis $D = (0, +\infty)$. Then x_o have the following properties.*

(1) $\lim_{\mathcal{I}\to+\infty} x_o(\mathcal{I}) = 0$,
(2) $\lim_{\mathcal{I}\to+\infty} \det(J) = +\infty$,
(3) $\lim_{\mathcal{I}\to+\infty} \det(H) = c > 0$,
(4) $\lim_{\mathcal{I}\to+\infty} x_o'(\mathcal{I}) = 0$.

In particular, x_o exhibits asymptotic infinitesimal homeostasis.

The proof proceed by a sequence of lemmas.

Lemma 3.2. $\lim_{\mathcal{I}\to+\infty} x_o(\mathcal{I}) = 0$.

Proof. From the equation for $\dot{x}_\iota$ in (3.48), we conclude that, for $\mathcal{I} > 0$, $x_\iota \neq 0$ and

$$x_\sigma = \frac{bx_\iota - \mathcal{I}}{ax_\iota} \tag{3.57}$$

Combining (3.57) with the equation for $\dot{x}_\sigma$ in (3.48), we obtain

$$x_o = \frac{(dx_\iota + e)(bx_\iota - \mathcal{I})}{acx_\iota - d(bx_\iota - \mathcal{I})} \tag{3.58}$$

From the equation of $\dot{x}_o$ in (3.48), the assumption that $j > 0$, implies that $x_o \neq 0$ and

$$x_\tau = \frac{b - ax_\sigma + lx_\iota}{h} - \frac{j}{hx_o} \tag{3.59}$$

Notice that, by (3.57), we have

$$b - ax_\sigma = b - a\frac{bx_\iota - \mathcal{I}}{ax_\iota} = \frac{\mathcal{I}}{x_\iota} \tag{3.60}$$

And therefore (3.59) is reduced to

$$x_\tau = \frac{\mathcal{I}}{hx_\iota} + \frac{lx_\iota}{h} - \frac{j}{hx_o} \tag{3.61}$$

From the equation of $\dot{x}_\tau$ in (3.48), the assumption that $j > 0$ implies that $x_\tau \neq 0$ and

$$x_o = \frac{b}{h} - \frac{(f+j)x_\tau + gj}{hx_\tau(x_\tau + g)} \tag{3.62}$$

In order to simplify the computations, let us suppose $bg = f + j$. Then, from (3.62) we obtain that

$$x_\tau^2(hx_o - b) + ghx_\tau x_o + gj = 0 \tag{3.63}$$

Now, combining (3.61) with (3.63) gives

$$(hx_o - b)(\mathcal{I}x_o + lx_\iota^2 x_o - jx_\iota)^2 + gh^2 x_o^3 x_\iota(\mathcal{I} + lx_\iota^2) = 0 \tag{3.64}$$

Writing x_o as a function of $\mathcal{I}$ and x_ι and inserting (3.58) into (3.64), we obtain

$$\begin{aligned} &[h(bdx_\iota^2 + ebx_\iota - e\mathcal{I}) - x_\iota(dh\mathcal{I} + bac) + bd(bx_\iota - \mathcal{I})] \\ &\times [\mathcal{I}(dx_\iota + e)(bx_\iota - \mathcal{I}) + lx_\iota^2(dx_\iota + e)(bx_\iota - \mathcal{I}) - acjx_\iota^2 + djx_\iota(bx_\iota - \mathcal{I})]^2 \\ &+ gh^2(dx_\iota + e)^3(bx_\iota - \mathcal{I})^3 x_\iota(\mathcal{I} + lx_\iota^2) = 0 \end{aligned} \tag{3.65}$$

Note that $\lim_{\mathcal{I}\to+\infty}(dh\mathcal{I} + bac) = \lim_{\mathcal{I}\to+\infty}(dh\mathcal{I})$ and hence, when limit $\mathcal{I} \to +\infty$, the polynomial equation described by (3.65) has the same solutions as the equation

$$\begin{aligned} &(bx_\iota - \mathcal{I})\{[h(dx_\iota + e) + bd][\mathcal{I}(dx_\iota + e)(bx_\iota - \mathcal{I}) + lx_\iota^2(dx_\iota + e)(bx_\iota - \mathcal{I}) \\ &- acjx_\iota^2 + djx_\iota(bx_\iota - \mathcal{I})]^2 + gh^2(dx_\iota + e)^3(bx_\iota - \mathcal{I})^2 x_\iota(\mathcal{I} + lx_\iota^2)\} = 0 \end{aligned} \tag{3.66}$$

One of the roots of (3.66) is given by $\lim_{\mathcal{I}\to+\infty} bx_\iota - \mathcal{I} = 0$ and so

$$\lim_{\mathcal{I}\to+\infty} x_\iota = +\infty \tag{3.67}$$

Finally, combining (3.67) with (3.57) and (3.58) gives

$$\begin{aligned} &\lim_{\mathcal{I}\to+\infty} x_\sigma = \lim_{\mathcal{I}\to+\infty} \frac{bx_\iota - \mathcal{I}}{ax_\iota} = 0 \\ &\lim_{\mathcal{I}\to+\infty} x_o = \lim_{\mathcal{I}\to+\infty} \frac{(dx_\iota + e)(bx_\iota - \mathcal{I})}{acx_\iota - d(bx_\iota - \mathcal{I})} = \lim_{\mathcal{I}\to+\infty} \frac{dx_\iota(bx_\iota - \mathcal{I})}{acx_\iota} = 0 \end{aligned} \tag{3.68}$$

This proves the Lemma. □

Lemma 3.3. $\lim_{\mathcal{I}\to+\infty} \det(J) = +\infty$.

Proof. We start by showing that $\lim_{\mathcal{I}\to+\infty} x_\tau = \alpha$. Suppose that $\lim_{\mathcal{I}\to+\infty} x_\tau = \pm\infty$ and consider the equation of $\dot{x}_\tau$ in (3.48) and take the limit $\mathcal{I} \to +\infty$ on both sides. Then

$$\begin{aligned} \lim_{\mathcal{I}\to+\infty} \dot{x}_\tau &= \lim_{\mathcal{I}\to+\infty} -bx_\tau + f\frac{x_\tau}{x_\tau + g} + hx_\tau x_o + j \\ &= \lim_{\mathcal{I}\to+\infty} x_\tau(-b + hx_o) + f\frac{x_\tau}{x_\tau + g} + j \\ &= \lim_{\mathcal{I}\to+\infty} -bx_\tau + f + j \\ &= \lim_{\mathcal{I}\to+\infty} -bx_\tau = \pm\infty \end{aligned} \tag{3.69}$$

Which is a contradiction since, for all $\mathcal{I} \in \mathbb{R}_+^*$, we have that $\dot{x}_\tau = 0$. Therefore, x_τ has a finite limit when $\mathcal{I} \to +\infty$. In particular, we have that $\lim_{\mathcal{I}\to+\infty} x_\tau x_o = 0$.

Now we consider the equation of $\dot{x}_\tau$ in (3.48) together with $\lim_{\mathcal{I}\to+\infty} x_\tau = \alpha$ and so

$$-b\alpha^2 + \alpha(-bg + f + j) + gj = 0 \tag{3.70}$$

As we assumed before that $f + j = bg$, Eq. (3.70) is reduced to

$$b\alpha^2 + gj = 0$$

Thus

$$\lim_{\mathcal{I}\to+\infty} x_\tau = \sqrt{\frac{gj}{b}} \tag{3.71}$$

In order to analyze the asymptotic behavior of J when $\mathcal{I} \to +\infty$ we use the limits computed before

$$\lim_{\mathcal{I}\to+\infty} J = \lim_{\mathcal{I}\to+\infty} \begin{pmatrix} -b & ax_\iota & 0 & 0 \\ 0 & -dx_\iota & 0 & c \\ 0 & 0 & \dfrac{bf - b^2g - bj - 2b\sqrt{bgj}}{gb + j + 2\sqrt{bgj}} & h\sqrt{\dfrac{gj}{b}} \\ 0 & 0 & 0 & -lx_\iota \end{pmatrix} \tag{3.72}$$

Therefore, the eigenvalues of $\lim_{\mathcal{I}\to+\infty} J$ are: $-b$, $\lim_{\mathcal{I}\to+\infty} -dx_\iota$, $\lim_{\mathcal{I}\to+\infty} -lx_\iota$ and

$$\frac{bf - b^2g - bj - 2b\sqrt{bgj}}{gb + j + 2\sqrt{bgj}}$$

By assumption $f + j = bg$ and so

$$f < bg \quad\Rightarrow\quad bf < b^2g \quad\Rightarrow\quad \frac{bf - b^2g - bj - 2b\sqrt{bgj}}{gb + j + 2\sqrt{bgj}} < 0$$

Hence, all the eigenvalues of $\lim_{\mathcal{I}\to+\infty} J$ are $-\infty$ or have negative real part. Therefore, the equilibrium is stable as $\mathcal{I} \to +\infty$. Finally, from the above property of the eigenvalues of $\lim_{\mathcal{I}\to+\infty} J$ and Eq. (3.67) it follows that

$$\lim_{\mathcal{I}\to+\infty} \det J = +\infty \tag{3.73}$$

This proves the Lemma. □

Lemma 3.4. $\lim_{\mathcal{I}\to+\infty} \det(H) = c > 0$.

Proof. As shown before

$$\lim_{\mathcal{I}\to+\infty} f_{\tau,\tau} = \frac{bf - b^2g - bj - 2b\sqrt{bgj}}{gb + j + 2\sqrt{bgj}} < 0 \tag{3.74}$$

Furthermore, from the equation of $\dot{x}_o$ in (3.48) we have that

$$\begin{aligned} \lim_{\mathcal{I}\to+\infty} ax_\sigma x_o - bx_o - lx_\iota x_o + hx_\tau x_o + j = 0 \\ \lim_{\mathcal{I}\to+\infty} -lx_\iota x_o + j = 0 \\ \lim_{\mathcal{I}\to+\infty} x_\iota x_o = \frac{j}{l} \end{aligned} \tag{3.75}$$

Thus, by combining (3.54) with (3.75), we get

$$\lim_{\mathcal{I}\to+\infty} f_{o,\sigma} f_{\sigma,\iota} - f_{o,\iota} f_{\sigma,\sigma} = \lim_{\mathcal{I}\to+\infty} -x_o[aex_\sigma + lex_\iota + lex_o + lf] = -ej \neq 0 \tag{3.76}$$

Finaly, from (3.74) and (3.76) it follows that

$$\lim_{\mathcal{I}\to+\infty} \det H = -ej \left(\frac{bf - b^2g - bj - 2b\sqrt{bgj}}{gb + j + 2\sqrt{bgj}} \right) = ej \left(\frac{b^2g + bj + 2b\sqrt{bgj} - bf}{gb + j + 2\sqrt{bgj}} \right) > 0$$

That is, $\lim_{\mathcal{I}\to+\infty} \det H = c$, a positive real number and this proves the Lemma. □

Lemma 3.5. $\lim_{\mathcal{I}\to+\infty} x_o'(\mathcal{I}) = 0$.

Proof. Now, from (3.73), the Cramer's Rule and the fact that $f_{\iota,\mathcal{I}} = 1$, we conclude that

$$\lim_{\mathcal{I}\to+\infty} \frac{dx_o}{d\mathcal{I}}(\mathcal{I}) = \lim_{\mathcal{I}\to+\infty} -f_{\iota,\mathcal{I}} \frac{\det H}{\det J} = 0$$

This proves the Lemma. □

It is interesting to observe that the conclusion that x_o exhibits *asymptotic infinitesimal homeostasis* follows from the divergence of $\det(J)$, which is, ultimately, caused by the divergence of one of the coordinate functions, namely, x_ι (see Eq. (3.67)). In other words, unlike in the usual infinitesimal homeostasis that is caused by the vanishing of one of the irreducible factors of $\det(H)$, asymptotic infinitesimal homeostasis cannot be attributed to an appendage or structural character. Furthermore, since the appendage or structural character is associated to subnetwork of the original network [12], the usual infinitesimal homeostasis is, in a certain sense, 'localized' at those subnetworks, whereas asymptotic infinitesimal homeostasis is a 'global' feature of the whole network.

4. Conclusion and outlook

In this paper we introduce two biologically motivated models and analyze the occurrence of *infinitesimal homeostasis* in those models [9,13,16]. The main goal to study these examples is to compare the 'generic' or 'model-independent' approach to infinitesimal homeostasis, based on the formalism of input–output networks [11,12], with the actual phenomena that occur in specific models.

The main goal of a 'model-independent' approach is to provide a catalog of the types of behavior that one should expect within a given class of models. This approach contrasts with the more common 'model-dependent' one, where specific model equations are considered. In biology, for example, precise laws are often not known [45]. Usually, model equations are obtained by a combination of a small set of biologically meaningful principles (law of mass action, Michaelis–Menten/Hill kinetics, etc.) and some *ad hoc* modeling choices, leading to very restricted functional forms of the differential equations. For example, chemical reaction systems typically obey the law of mass action and the component functions of the differential equations are given by polynomial functions [46].

Therefore, we have showed here that the lack of genericity of the specific model equations lead to very 'unexpected' phenomena. We analyzed both examples by first employing a 'model-independent' approach and then considering the specific model equations. We illustrated the phenomena by numerical simulation and afterwards provided rigorous arguments for some of them. As we could see, the proofs are elementary but quite involved and mostly relying on *ad hoc* arguments.

In the first case, the *copper intracellular regulation* model, we showed that, from the 'model-independent' point of view there are two 'mechanisms' that can cause infinitesimal homeostasis, but in the specific model, only one of them can occur. We are able to show that the singularity type of the infinitesimal homeostasis point, when it exists, is the *simple homeostasis* type. Moreover, we uncover, by numerical simulation, a phenomena in which, under variation of second parameter, the homeostasis point and a nearby limit point 'collide' and 'annihilate' each other, leading to a configuration where there is no infinitesimal homeostasis at all. Finally, both situations (simple homeostasis and no infinitesimal homeostasis) occur 'robustly' in the specific model, that is, they occur for an open set of parameters. Several questions about this model were not addressed, such as

- When there is no infinitesimal homeostasis, numerical simulation suggests that the input–output function is defined over $D = [0, +\infty)$. Is that true in a robust way? Is the input–output function unbounded or bounded on D? In the last case is it asymptotically homeostatic ? See Appendix A.
- When there exists a simple homeostasis point, numerical simulation suggests that there is a 'partner' limit point. Is that true in a robust way? Is the collision of both points under variation of a second parameter a robust phenomena?

In the second case, the *self immune recognition* model, we showed that from the 'model-independent' point of view there are two 'mechanisms' that can cause infinitesimal homeostasis, but in the specific model, none of them can occur. We are able to show that when a certain parameter of the model vanishes, the input–output function is the zero constant function. This kind of phenomenon is called *perfect homeostasis*. In the control theory literature this is condition is called *perfect adaptation* [18–20,47–51]. Usually, the functional form of the differential equations must be strongly constrained in order to get a constant input–output function. In fact, this is the case in the self immune recognition model, when the above mentioned parameter vanishes (see Eq. (3.53)). However, when the above mentioned parameter is non-zero the model exhibits no infinitesimal homeostasis at all. In this case, the numerical simulation suggested that the input–output function x_o was defined on $D = [0, +\infty)$ and $x_o \to 0$ as the input parameter $\mathcal{I} \to +\infty$. In fact, it seems that a stronger property is true: that the input–output function exhibits *asymptotic infinitesimal homeostasis*, that is, in addition to the above, one has that $x_o' \to 0$, as well (see Appendix A). We were able to show that this is, indeed, true, assuming that the input–output function is defined on $D = [0, +\infty)$. This last condition is rather difficult to address in our context, since the input–output function is often shown to be defined by application of the implicit function theorem. Therefore, we can only assert that the input–output function is well-defined on a bounded interval. Perhaps some 'global' implicit function theorem could be helpful in this case [52].

CRediT authorship contribution statement

Pedro P.A. Cardoso de Andrade: Writing – review & editing, Writing – original draft, Investigation, Conceptualization. **João L.O. Madeira:** Writing – review & editing, Writing – original draft, Investigation, Formal analysis, Conceptualization. **Fernando Antoneli:** Writing – review & editing, Writing – original draft, Supervision, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Fernando Martins Antoneli Junior reports financial support was provided by Federal University of Sao Paulo. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The research of FA was supported by Fundação de Amparo à Pesquisa do Estado de São Paulo (FAPESP) grants 2019/12247-7 and 2019/21181-0.

Appendix A. Infinitesimal homeostasis at infnity

In this section we extend the theory of [9,12,15,16] to the case where the input–output function satisfies the near-perfect homeostasis condition on a semi-infinite interval and the infinitesimal homeostasis occurs at the 'infinite end'.

A.1. Asymptotic infinitesimal homeostasis

Consider a network $\mathcal{G}$ such that the associated input–output function z is defined on a semi-infinite interval $D = (\mathcal{I}_0, +\infty)$.

Theorem A.1. *Let $z : D \to \mathbb{R}$ be a smooth function, with $D = (\mathcal{I}_0, +\infty)$. Suppose that z satisfies the near-perfect homeostasis condition over D: for all $\mathcal{I} \in D, z(\mathcal{I}) \in (z(\mathcal{I}_{sp}) - \delta, z(\mathcal{I}_{sp}) + \delta)$, for some $\mathcal{I}_{sp} \in D$ and fixed $\delta > 0$. Then, at least one of the following statements is true:*

(i) There exists $\mathcal{I}_c \in D$ such that $z'(\mathcal{I}_c) = 0$,
(ii) There exists an increasing sequence $(\mathcal{I}_n)_{n\geqslant 1} \subset D$ satisfying

$$\lim_{n\to\infty} \mathcal{I}_n = +\infty \qquad \text{and} \qquad \lim_{n\to\infty} z'(\mathcal{I}_n) = 0.$$

In particular, if z' is a monotonic function, then $\lim_{\mathcal{I}\to+\infty} z'(\mathcal{I}) = 0$.

Proof. Suppose there exists $\mathcal{I}_1, \mathcal{I}_2 \in D$ such that $z'(\mathcal{I}_1) \cdot z'(\mathcal{I}_2) \leqslant 0$. If $z'(\mathcal{I}_1) \cdot z'(\mathcal{I}_2) = 0$, then $z'(\mathcal{I}_1) = 0$ or $z'(\mathcal{I}_2) = 0$, and thus (i) is true. On the other hand, if $z'(\mathcal{I}_1) \cdot z'(\mathcal{I}_2) < 0$, then $z'(\mathcal{I}_1) > 0$ and $z'(\mathcal{I}_2) < 0$ or $z'(\mathcal{I}_1) < 0$ and $z'(\mathcal{I}_2) > 0$. In both cases, by the mean value theorem, there exists $\mathcal{I}^\star \in (\mathcal{I}_1, \mathcal{I}_2)$ such that $z'(\mathcal{I}^\star) = 0$, and thus (i) is true.

Now suppose that for all $\mathcal{I}_1, \mathcal{I}_2 \in D$, $z'(\mathcal{I}_1) \cdot z'(\mathcal{I}_2) > 0$. This means that $z'(\mathcal{I})$ is either positive or negative over D. Let us consider the case where $z'(\mathcal{I})$ is positive over D (the other case is analogous). Since $z'(D)$ is bounded $\inf z'(D) \geq 0$. Consider the dyadic sequence $\mathcal{J}_n = \sum_{m=0}^{n} 2^m$, for $n \geqslant 0$, and define a family of consecutive disjoint intervals $(J_n)_{n\geqslant 1}$ contained in D, of length 2^n, by $J_n = \big(\mathcal{I}_0 + \mathcal{J}_{n-1}, \mathcal{I}_0 + \mathcal{J}_n\big)$. Hence, one can write

$$2^n \inf z'(J_n) = \int_{\mathcal{I}_0+\mathcal{J}_{n-1}}^{\mathcal{I}_0+\mathcal{J}_n} \inf z'(J_n)\, \mathrm{d}\,\mathcal{I} \leqslant \int_{\mathcal{I}_0+\mathcal{J}_{n-1}}^{\mathcal{I}_0+\mathcal{J}_{n-1}} z'(\mathcal{I})\, \mathrm{d}\,\mathcal{I} \leqslant 2\delta$$

and so

$$\inf z'(J_n) \leqslant \frac{\delta}{2^{n-1}}$$

Therefore, there exists $\mathcal{I}_n \in J_n$, for $n \geqslant 1$, such that

$$0 < z'(\mathcal{I}_n) \leqslant \inf z'(J_n) + \frac{\delta}{2^{n-1}} \leqslant \frac{2\delta}{2^{n-1}} \tag{A.77}$$

It is clear that $(\mathcal{I}_n)_{n\geqslant 1}$ is an increasing sequence with $\lim_{n\to\infty} \mathcal{I}_n = \infty$. By (A.77), we conclude that

$$\lim_{n\to\infty} z'(\mathcal{I}_n) = 0$$

and therefore (ii) is true. Finally, it is obvious that, if z' is a monotonic function, then $\lim_{\mathcal{I}\to\infty} z'(\mathcal{I}) = 0$. □

Note that the near perfect homeostasis condition is equivalent to the condition that $|z|$ is *bounded* on D, namely, there exists som $M > 0$ such that $|z| \leqslant M$ on D.

Definition A.1. An input–output function $z : D \to \mathbb{R}$, with $D = (\mathcal{I}_0, +\infty)$, exhibits *asymptotic infinitesimal homeostasis* if it is bounded on D and

$$\lim_{\mathcal{I}\to\infty} z'(\mathcal{I}) = 0.$$

In particular, Theorem A.1 is just a criterion for the vanishing of the derivative of a function on infinity.

Corollary A.2. *Let $z : D \to \mathbb{R}$, with $D = (\mathcal{I}_0, +\infty)$ be a smooth function. If z is bounded on D and z is monotonic on D, then $\lim_{\mathcal{I}\to+\infty} z'(\mathcal{I}) = 0$.*

If z is bounded on D and z and is monotonic on D then

$$\lim_{\mathcal{I}\to+\infty} z(\mathcal{I}) = c \tag{A.78}$$

for some finite $c \in \mathbb{R}$. However, the converse is not necessarily true. The existence of the limit (A.78) implies that $|z| \leqslant M$ on some $D' = (\mathcal{I}_0', +\infty)$ with $\mathcal{I}_0' > \mathcal{I}_0$, and so $D' \subset D$, but z may have critical points in D (and D'). For example, $z(x) = \sin(x)/x$.

Moreover, the existence of the limit (A.78) is necessary to ensure that the condition of the derivative vanishing at infinity really captures the notion of 'critical point at infinity' [53]. For example, consider $z(x) = \log(x)$ and $z'(x) = 1/x$ on $D = (0, +\infty)$. The derivative $z'(x)$ vanishes at infinity but z is unbounded on D.

The existence of the limit (A.78) is a natural necessary condition for a 'higher order singularity at infinity', as in the next classical result.

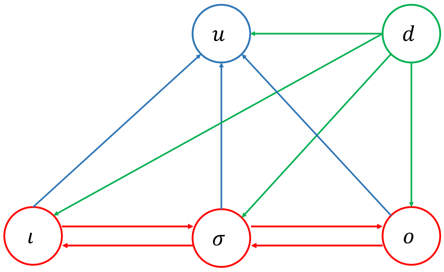

Fig. 8. Partition of nodes of $\mathcal{G}$. Subnetwork in red is the core network $\mathcal{G}_c$. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Theorem A.3 (*Hadamard's Lemma [54]*). *Let $z : D \to \mathbb{R}$, with $D = (\mathcal{I}_0, +\infty)$ be a smooth function. If the first n derivatives $z', \dots, z^{(n)}$ of z are bounded on D, then $\lim_{\mathcal{I}\to+\infty} z(\mathcal{I}) = c$, for some finite $c \in \mathbb{R}$ implies that*

$$\lim_{\mathcal{I}\to+\infty} z^{(j)}(\mathcal{I}) = 0$$

for all $1 \le j \le n$.

Gabriel and Berrut [55] provide a nice review on the question of whether the derivative of a function vanishes at infinity and they arrive at a necessary and sufficient condition.

Theorem A.4 (*Gabriel & Berrut [55]*). *Let $z : D \to \mathbb{R}$, with $D = (\mathcal{I}_0, +\infty)$ be a C^1 function such that $\lim_{\mathcal{I}\to+\infty} z(\mathcal{I}) = c$, for some finite $c \in \mathbb{R}$. Then $\lim_{\mathcal{I}\to+\infty} z'(\mathcal{I}) = 0$ if and only if z' is uniformly continuous on D.*

A.2. Core network reduction for asymptotic infinitesimal homeostasis

Golubitsky et al. [12] have shown that in order to analyze if an input–output network exhibits infinitesimal homeostasis, it is enough to study an associated *core network*, i.e., a network in which every node is downstream from the input node ι and upstream from the output node o. We will show that this theorem extends to the case of asymptotic infinitesimal homeostasis.

Let $\mathcal{G}$ be an input–output network with input node ι, output node o and regulatory nodes ρ. Partition the nodes of $\mathcal{G}$ three types:

- those nodes σ that are both upstream from o and downstream from ι,
- those nodes d that are not downstream from ι
- those nodes u which are downstream from ι, but not upstream from o

Fig. 8 exhibits this partition of regulatory nodes of $\mathcal{G}$.

The generic system of ODEs associated to the original network $\mathcal{G}$ is given by

$$\begin{aligned}
\dot{x}_\iota &= f_\iota(x_\iota, x_\sigma, x_d, x_u, x_o, \mathcal{I})\\
\dot{x}_\sigma &= f_\sigma(x_\iota, x_\sigma, x_d, x_u, x_o)\\
\dot{x}_d &= f_d(x_\iota, x_\sigma, x_d, x_u, x_o)\\
\dot{x}_u &= f_u(x_\iota, x_\sigma, x_d, x_u, x_o)\\
\dot{x}_o &= f_o(x_\iota, x_\sigma, x_d, x_u, x_o)
\end{aligned} \tag{A.79}$$

The reduced systems of ODEs associated to the core network $\mathcal{G}_c$ obtained from (A.79) is given by

$$\begin{aligned}
\dot{x}_\iota &= f_\iota(x_\iota, x_\sigma, x_d, x_o, \mathcal{I})\\
\dot{x}_\sigma &= f_\sigma(x_\iota, x_\sigma, x_d, x_o)\\
\dot{x}_d &= f_d(x_d)\\
\dot{x}_u &= f_u(x_\iota, x_\sigma, x_d, x_u, x_o)\\
\dot{x}_o &= f_o(x_\iota, x_\sigma, x_d, x_o)
\end{aligned} \tag{A.80}$$

Theorem A.5. *Let $x_o(\mathcal{I})$ be the input–output function of the admissible system (A.79) and let $x_o^c(\mathcal{I})$ be the input–output function of the associated core admissible system (A.80). Consider that both functions are defined in the semi-infinite interval $D = (\mathcal{I}_0, +\infty)$. Then, the input–output function $x_o^c(\mathcal{I})$ associated to the core subnetwork $\mathcal{G}_c$ exhibits asymptotic infinitesimal homeostasis if and only if the input–output function $x_o(\mathcal{I})$ associated to the original network $\mathcal{G}$ exhibits asymptotic infinitesimal homeostasis.*

Proof. The Jacobian J the original network is

$$J = \begin{pmatrix} f_{\iota,x_\iota} & f_{\iota,x_\sigma} & f_{\iota,x_d} & 0 & f_{\iota,x_o} \\ f_{\sigma,x_\iota} & f_{\sigma,x_\sigma} & f_{\sigma,x_d} & 0 & f_{\sigma,x_o} \\ 0 & 0 & f_{d,x_d} & 0 & 0 \\ f_{u,x_\iota} & f_{u,x_\sigma} & f_{u,x_d} & f_{u,x_u} & f_{u,x_o} \\ f_{o,x_\iota} & f_{o,x_\sigma} & f_{o,x_d} & 0 & f_{o,x_o} \end{pmatrix} \tag{A.81}$$

and the corresponding homeostasis matrix H is

$$H = \begin{pmatrix} f_{\sigma,x_\iota} & f_{\sigma,x_\sigma} & f_{\sigma,x_d} & 0 \\ 0 & 0 & f_{d,x_d} & 0 \\ f_{u,x_\iota} & f_{u,x_\sigma} & f_{u,x_d} & f_{u,x_u} \\ f_{o,x_\iota} & f_{o,x_\sigma} & f_{o,x_d} & 0 \end{pmatrix} \tag{A.82}$$

On the other hand, the Jacobian J^c and the homeostasis matrix H^c of the core network are, respectively:

$$J^c = \begin{pmatrix} f_{\iota,x_\iota} & f_{\iota,x_\sigma} & f_{\iota,x_o} \\ f_{\sigma,x_\iota} & f_{\sigma,x_\sigma} & f_{\sigma,x_o} \\ f_{o,x_\iota} & f_{o,x_\sigma} & f_{o,x_o} \end{pmatrix} \quad \text{and} \quad H^c = \begin{pmatrix} f_{\sigma,x_\iota} & f_{\sigma,x_\sigma} \\ f_{o,x_\iota} & f_{o,x_\sigma} \end{pmatrix} \tag{A.83}$$

Then we can compute

$$\begin{aligned} \det H &= (-1)^{k_H} \det(f_{d,x_d}) \det(f_{u,x_u}) \det H^c \\ \det J &= (-1)^{k_J} \det(f_{d,x_d}) \det(f_{u,x_u}) \det J^c \end{aligned} \tag{A.84}$$

Now, for all $\mathcal{I} \in D$, J and J^c must have eigenvalues with negative real part. As the eigenvalues of f_{d,x_d} and of f_{u,x_u} are also eigenvalues of J, we conclude that

$$\det(f_{d,x_d}) \cdot \det(f_{u,x_u}) \neq 0 \tag{A.85}$$

Therefore

$$\lim_{\mathcal{I}\to\infty} \frac{\det H}{\det J} = \lim_{\mathcal{I}\to\infty} \frac{(-1)^{k_H} \det(f_{d,x_d}) \det(f_{u,x_u}) \det H^c}{(-1)^{k_J} \det(f_{d,x_d}) \det(f_{u,x_u}) \det J^c} = (-1)^k \lim_{\mathcal{I}\to\infty} \frac{\det H^c}{\det J^c} \tag{A.86}$$

which concludes the proof. □

Data availability

No data was used for the research described in the article.

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